Error Analysis of isiZulu Street Name Pronunciation in Existing Systems

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1. Introduction

1.1 Background on NLP and speech technologies

Natural Language Processing (NLP) and speech technologies have significantly transformed human-computer interactions, fostering more natural communication between users and digital systems (Manaris, 1998). Human Language Technology (HLT) systems enable computers to process and generate human language, supporting applications such as dialogue systems, virtual assistants and GPS navigation tools (Nthite & Tsoeu, 2020). These technologies rely on Automatic Speech Recognition (ASR) and Text-to-Speech (TTS) to interpret and produce speech, their performance has improved considerably with advances in deep learning and the availability of large-scale datasets (Biswas et al., 2018).

Despite these advancements, the development of NLP and speech technologies has been uneven, with high-resourced languages such as English and other European languages dominating research and data availability, while African languages remain underrepresented (Sefara & Manamela, 2019). This imbalance has resulted in systems such as GPS navigation performing well in widely supported languages but struggling to process local languages like isiZulu (Bhushan et al., 2025).

According to the 2022 Census, isiZulu, is one of the most widely spoken languages in South Africa, spoken by 24.4% of the country's total population of approximately 62 million (Statistics South Africa, 2022). IsiZulu presents unique challenges for speech technologies because it is characterised by a rich phonetic structure, complex sound patterns and agglutinative morphology, which make accurate pronunciation difficult for standard computational models (Keet & Khumalo, 2017). These challenges are evident in real-world applications like GPS navigation systems, including Google Maps and Waze, where isiZulu street names are frequently mispronounced. As a result, human recordings are often used to support or validate pronunciation accuracy, highlighting the limitations of current speech technologies (Demin et al., 2025).

1.2 The importance of African Language technologies

African language technologies extend beyond technical performance and enters into broader issues of accessibility, digital inclusion, and linguistic representation. African languages account for a significant portion of global linguistic diversity, yet they remain severely underrepresented in modern NLP and speech technologies, creating barriers for millions of speakers who rely on indigenous languages in their everyday lives (Agic &

Vulic, 2020; Imam et al., 2025). As digital systems become increasingly integrated into communication, navigation and public services, ensuring support for African languages is essential for promoting equitable access to technology.

For isiZulu speakers, inaccuracies in systems such as GPS navigation tools and text-to-speech applications are not merely a technical limitation but also affect usability and effective communication. Mispronunciation of street names and locations can reduce user trust in these systems and potentially lead to navigation errors or safety risks (Nthite & Tsoeu, 2020). These challenges are evident in speech technologies that rely on generalised multilingual datasets which often fail to capture the phonetic and morphological characteristics of low-resource African languages (Diack et al., 2026).

The need for localised language technologies has also become increasingly important within South African research communities. Recent initiatives such as MzanziLM, developed at the University of Cape Town (UCT), represent emerging efforts to build language models capable of addressing the linguistic complexities of South African languages through locally relevant datasets and modelling approaches (Lombard et al., 2026). Such initiatives highlight the importance of developing systems that account for the phonetic nuances, agglutinative structures, and unique sound patterns found in languages such as isiZulu.

Furthermore, supporting African language technologies contribute to preserving cultural identity and improving representation within the digital landscape (Mlambo & Matfunjwa, 2024). Current global speech systems are largely optimised for high-resource languages and usually struggle with features unique to isiZulu, including click consonants and complex word formations (Sibizo, 2025). Addressing these limitations requires speech technologies that are trained and evaluated using locally grounded linguistic data instead of generalised datasets (Lokeshkiran & Karthikeyan, 2025).

This study will therefore focus on analysing pronunciation errors in existing speech-based systems by comparing system-generated pronunciations with ground truth recordings collected from native isiZulu speakers. Through this analysis, the study aims to identify recurring pronunciation error patterns and contribute toward improving speech technologies for isiZulu.

2. Problem statement

Current speech-based navigation and text-to-speech systems are primarily designed for English and other well-resourced languages, resulting in inaccurate pronunciation of isiZulu street names (Sefara & Manamela, 2019; Liu, Yang & Qu, 2024). The inaccuracies arise from many factors, including limitations in phonetic modelling, insufficient or biased training data, and the inability of existing systems to effectively handle the agglutinative structure and phonetic complexity of isiZulu (Koffi, 2020; Diack *et al.*, 2026). As a result, users frequently experience mispronunciations that affect usability, reduce trust in digital systems, and limit accessibility on new technologies that could improve and empower the lives of native speakers (Nthite & Tsoeu, 2020).

Despite ongoing advancements in speech technologies, isiZulu and other African languages remain underrepresented in both datasets and system design. There is therefore need for error analysis that is structured to systematically identify these pronunciation issues, understand their underlying causes, and contribute towards the development of more inclusive and accurate speech technologies.

3. Research Questions

Main Research Question

To what extent do existing speech-based systems accurately pronounce isiZulu street names?

This question focuses on evaluating the overall performance of current systems in handling isiZulu pronunciation, particularly within applications such as navigation and text-to-speech systems.

Research Sub Questions

1. **What types of pronunciation errors occur in these systems?**

This question aims to identify and classify the different categories of errors, such as phoneme substitutions, stress or tone errors, and issues related to agglutinative structures.

2. **What linguistic and computational factors contribute to these errors?**

This question explores the underlying causes of pronunciation issues, including phonetic complexity, limited training data and model limitations.

3. **How does pronunciation accuracy vary across different speech-based systems when handling isiZulu street names?**

This question focuses on evaluating variations in performance across systems such as GPS navigation tools and text-to-speech applications.

4. **How can the identified errors inform improvement in speech technologies for African languages?**

This question aims to link the findings of the study to practical recommendations for improving speech systems and supporting low-resource languages.

4. Research objectives

The objectives of this study are:

1. To collect a dataset of isiZulu street names and their corresponding pronunciations.
2. To obtain ground truth pronunciations from native isiZulu speakers across different dialectal backgrounds within the KwaZulu-Natal region.
3. To evaluate how existing speech-based systems pronounce isiZulu street names.
4. To identify and categorise pronunciation errors produced by these systems.
5. To analyse the possible phonetic and linguistic causes of these pronunciation errors.
6. To provide insights that may contribute towards improving speech technologies for isiZulu.
7. To investigate the development and training of a speech model using the collected pronunciation dataset in order to improve the pronunciation accuracy of isiZulu street names in South African navigation systems.

5. Initial dataset description

5.1 Data Source of data

The initial dataset consisted of verified isiZulu street names collected from the Umlazi and eThekwini regions in KwaZulu-Natal, South Africa. These street names were obtained through personal observation, local familiarity and everyday navigation experience as well as through online mapping platforms such as Apple Maps. All collected names were verified using a web-based data mining tool for Open Street Map (OSM) called overpass turbo to ensure accuracy and consistency before integration into the recording platform.

5.2 Data collection approach

The initial phase of data collection focused on creating a pronunciation dataset for evaluating how existing navigation systems pronounce isiZulu street names. A total of 150 verified isiZulu street names were manually collected, the names comprised of 50 street names from Umlazi township followed by 100 additional street names from eThekwini Metropolitan Municipality.

The verified street names were then submitted to a recording platform called NaVoice, where multiple native isiZulu speakers contributed reference pronunciations. These recordings serve as the ground truth pronunciations for the study. Approximately 50 corresponding navigation pronunciations were collected from Apple Maps to provide corresponding navigation pronunciations for comparison against the human recordings.

The dataset metadata is structured to include the street name, language, system output, and corresponding ground truth pronunciation. This dataset will then continue to expand toward the long-term objective of building a 100-hour isiZulu speech corpus that will provide sufficient phonetic and pronunciation variabilities suitable for training a speech model that will be capable of producing accurate pronunciations of isiZulu street names withing navigation systems.

The following metadata were maintained for each recording:

- Street name
- Language (isiZulu)
- Geographic location (Umlazi or eThekwini)
- Province (KwaZulu-Natal)
- System pronunciation (Apple Maps)
- Ground truth pronunciation (NaVoice)
- Contributor ID
- Audio format (WAV)
- Sampling rate (16 kHz)
- Recording device
- Speaker category
- Recording date

The dataset was structured to facilitate pronunciation comparison, error analysis and subsequent expansion into a larger speech corpus for future speech model development.

5.3 Challenges encountered

Several challenges were encountered during the initial data collection process. Firstly, the availability of isiZulu street names in digital mapping platforms is uneven, particularly in less documented areas (Adebara, 2025). Secondly, obtaining accurate and consistent ground truth pronunciation requires input from multiple native speakers to account for dialectal variation, which needs extra efforts to reach those people considering that some individuals may not be interested in participating due to the time required for recording (Way With Words Services, 2024). Thirdly, technical challenges emerged when attempting to align human recordings with system generated outputs due to differences in audio formats, and pronunciation representation. In addition, building a diverse and reliable voice dataset required consideration of contributor demographics such gender and age group, however, participation levels were inconsistent because some individuals were reluctant to register on the platform, which in turn introduced further complications in the process of data collection and representation (Babirye et al., 2022). These challenges highlight the complexity of developing reliable datasets for African language technologies.

6. Literature Review

Natural Language Processing (NLP) and speech technologies have transformed human-computer interactions. While deep learning has significantly improved speech synthesis and recognition for high-resource languages like English and Mandarin (Manaris, 1998; Jurafsky & Martin, 2026), there is still a huge disparity that remains for low-resource languages. African languages, particularly isiZulu, normally face limited support because there is a massive shortage of annotated corpora, pronunciation lexicons, and language-specific computational resources (Besacier et al., 2014). This disparity is particularly evident in navigation systems such as Apple Maps, Google Maps and Waze, where the mispronunciation of these indigenous street names diminishes intelligibility and further undermines user trust and how they perceive these GPS tools.

Existing literature on Natural Language Processing (NLP), speech technologies, African language technologies, and pronunciation evaluation provides the theoretical foundation for this study. Previous research on Text-to-Speech (TTS), Automatic Speech Recognition (ASR), pronunciation modelling, and speech evaluation is reviewed with particular emphasis on low-resource African languages such as isiZulu. The review further examines the linguistic characteristics that make isiZulu challenging for speech technologies, discusses pronunciation errors reported in navigation systems, and considers established approaches for pronunciation error analysis. Collectively, these studies identify the existing research gap and provide the theoretical basis for the methodology adopted in this study.

6.1 Natural Language Processing and Speech Technologies

Natural language processing (NLP) and speech technologies are now essential parts of today's intelligent systems that allow communication between humans and computers through spoken and written language (Manaris, 1998). Technologies such as Automatic Speech Recognition (ASR) and Text-to-Speech (TTS) systems are widely applied in virtual assistants, automated transcription services, accessibility tools, and GPS navigation systems including Google Maps and Waze (Khanam, et al., 2022). Khanam et al (2022), further alluded that speech synthesis and accuracy recognition has significantly improved in high-resource languages such as English and Mandarin through recent advances in deep learning, neural networks and transformer-based architectures.

The evolution of TTS has moved from robotic concatenative synthesis toward neural architectures like Tacotron, an autoregressive acoustic model that converts text into Mel spectrograms and Variational Inference Text-to-Speech (VITS), which generate high-fidelity audio (Kim, Kong & Son, 2021). In spite of these advancements, the performance of TTS tools remains highly dependent on the availability of larger annotated datasets and language-specific resources. Languages such as isiZulu with limited digital resources experience lower system accuracy and poor pronunciation modelling (Gunduz, et al., 2024). While English datasets contain thousands of hours of audio, many African languages only have a fraction of this data, this leads to a digital divide in speech quality (Besacier et al., 2014; Pratap et al., 2023). Besacier et al (2014), further alluded that low-resource languages continue to face huge challenges in speech technology development due to insufficient corpora, limited phonetic resources, and inadequate computational modelling approaches.

Among the practical applications of modern Text-to-Speech technology, navigation systems represent one of the most widely used real-world implementations (Jwo, Biswal & Mir, 2023). Platforms such as Apple Maps, Google Maps and Waze generate spoken driving instructions by converting textual street names into synthetic speech (Zhao, Haffari & Shareghi, 1975). The quality of these spoken directions depends on the availability of language-specific pronunciation resources, making indigenous African street names particularly challenging for multilingual speech models (Olatunji et al., 2023). This challenge motivates the present study, which investigates the pronunciation accuracy of isiZulu street names generated by Apple Maps.

6.2 African Language Technologies and Low-Resource Languages

A language is classified as "low-resource" when it lacks the digital data required for robust NLP applications. Scarcity of textual corpora and pronunciation lexicons makes developing reliable speech systems for African languages difficult (Besacier *et al.*, 2014).

Despite being South Africa’s most widely spoken home language, isiZulu remains underrepresented in commercial technologies. In recent initiatives such as the Masakhane community and the Mozilla Common Voice, have attempted to address this by collecting multilingual datasets (Ardila et al., 2020; Nekoto et al., 2020). However, these datasets often focus on general vocabulary rather that specialised geographic entities such as street names. Research therefore, suggests that while self-supervised learning is able to transfer knowledge between languages, language specific datasets remain essential for accurate pronunciation (Pratap *et al.*, 2023).

One approach that would help address the shortage of speech resources is the use of community-based recording platforms which enable native speakers to contribute pronunciation data. In this study, the NaVoice platform was selected as the primary source to collect multiple native-speaker pronunciations of isiZulu street names. Such collaborative data collection not only increase the quantity of available speech resources but it would also capture natural pronunciation variation which can subsequently be used to establish a representative consensus ground truth.

Speech pronunciation naturally varies between native speakers because of differences in speaking rate, pitch, accent and recording conditions. Consequently, relying on a single recording as the reference pronunciation may introduce speaker-specific bias into the evaluation process. Recent speech evaluation research therefore recommends constructing a consensus ground truth by combining multiple native-speaker recordings to establish a more representative pronunciation reference, this approach improves the reliability of pronunciation evaluation while reducing the influence of individual speaker variation (Nguyen et al., 2025).

6.3 Linguistic Challenges in isiZulu Speech Processing

IsiZulu, a Bantu language that belongs to the Nguni group of languages, presents unique challenges for speech processing because of its agglutinative morphology and complex phonology (Keet & Khumalo, 2016).

6.3.1 Agglutination and Morphology

According to Keet and Khumalo (2017), multiple morphemes in isiZulu are combined to make a single word. This requires speech models to be able to correctly identify morpheme boundaries so that they can maintain natural coarticulation. Systems trained on non-agglutinative languages like English often segment words incorrectly, this results in unnatural pauses or syllable boundaries (Shezi & Reddy, 2020).

6.3.2 Click Consonants and Phonetic Complexity

IsiZulu features a distinctive phonological inventory that include click consonants. Produced through a lingual ingressive airstream mechanism, clicks are represented orthographically by 'c' (dental), 'q' (palato-alveolar), and 'x' (lateral). These sounds present a dual-articulation mechanism that involves complex spatial and temporal adjustments (Vilakati & Kimberly D, 2010).

Phonetic research by Thomas Vilakati (2010) further highlights that clicks exhibit extensive coarticulation with surrounding vowels and consonants, which challenges traditional discrete phonetic modelling. For example, the dorsal closure of a click may shift based on the position of the vowel that follows. This coarticulatory complexity is often overlooked in commercial TTS systems, which may simplify the click articulation or substitute it with more familiar velar stops (Roux, 2007). Click production involves lowering the tongue centre to achieve rarefaction, a process that has so many variations across the three click types. Consequently, multilingual models trained on Germanic languages (which share sounds with Bantu languages such as isiZulu) often lack the acoustic resolution to capture these nuances, resulting in severe pronunciation distortions in navigation instructions (Van der Merwe & le Roux, 2014).

6.4 Speech Technology in Navigation Systems

Navigation applications such as Apple Maps and Google Maps integrate GPS with TTS to provide turn-by-turn instructions. These systems often rely on Grapheme-to-Phoneme (G2P) prediction rather than dictionaries for street names (Taylor, 2009). When they encounter indigenous names like '*Mgaga Road'*, these systems may apply English pronunciation rules, leading to errors in phoneme substitution, incorrect syllable stress, tone misplacement and difficulties handling agglutinative word structures (Koffi, 2020). Such errors reduce intelligibility, negatively affect user trust, and may create confusion during navigation.

Navigation systems such as Google Maps and Waze often rely on pronunciation models optimised for English phonological rules, resulting in distorted pronunciations of African street names and locations (Bhushan *et al.*, 2025). Accurate mapping data does not guarantee accurate pronunciation. While geographic databases are frequently updated, the associated phonetic information for local place names is often absent (Goodchild, 2007). This creates a critical research gap: how to evaluate and improve the pronunciation of isiZulu street names that are poorly represented in standard training sets.

Although commercial navigation systems provide increasingly natural synthetic speech, evaluating every generated pronunciation manually becomes impractical as the number of recordings increases. Consequently, recent pronunciation evaluation studies advocate the use of automated acoustic analysis techniques capable of comparing system-generated pronunciations with multiple native-speaker recordings using objective acoustic measurements (Korzekwa et al., 2022). This approach enables reproducible large-scale evaluation while reducing reliance on subjective listening alone.

6.5 Error Analysis and Evaluation Approaches in Speech Technologies

Refining speech systems requires systematic error detection. In recent years, researchers have shifted toward automated frameworks that can handle large datasets using specialised computational tools.

6.5.1 Batch Processing and Automated Metrics

Modern evaluation frameworks utilise batch processing scripts that are normally developed in languages like Python, these scripts automate the comparison of system-generated audio against ground truth recordings. This approach enables the analysis of the entire datasets, for instance, hundreds of street names rather than isolated samples. Automated pipelines can incorporate error handling to identify missing files, empty recordings, or processing failures, which in turn ensures data integrity across large scale experiments (Rudolph Van Niekerk, 2009).

Central to these frameworks are quantitative metrics such as the Pronunciation Accuracy Score (PAS), which integrates Mel-Frequency Cepstral Coefficients (MFCCs) for spectral analysis and Dynamic Time Warping (DTW) for temporal alignment (Davis & Mermelstein, 1980). By calculating duration differences and pitch deviations automatically, these scripts provide a high-resolution view of system performance across diverse linguistic categories.

In addition to quantitative measurements, acoustic visualisations play an important role in pronunciation analytics that cannot always be observed through numerical metrics alone. Davis and Mermestein (1980), further alluded that waveform allow researchers to identify silence, speech onset and duration differences, while spectrograms display the distribution of acoustic energy across frequencies over time. Mel Frequency Cepstral Coefficients (MFCCs) provide a compact representation of the spectral characteristic of speech and are widely used when comparing pronunciations computationally. Together, these visualisation techniques complement objective evaluation metrics, allowing researchers to interpret pronunciation differences more comprehensively.

6.5.2 Data Visualisation and Heatmaps

To interpret large volumes of acoustic data, researchers employ visual analytics. Heatmaps are increasingly used as acoustic similarity matrices to quantify relationships between produced speech segments and target phonetic models (Rudolph Van Niekerk, 2009). These similarity heatmaps would allow for the visualisation of error distributions across different phonetic clusters, identifying whether specific types of sounds such as clicks or nasal clusters are consistently mispronounced across the dataset (Kraus et at., 2020). Other common visualisations include PAS score distributions, processing status summaries, and scatter plots comparing DTW scores against pitch variability, which help in identifying the worst-performing street names for further qualitative analysis.

An error taxonomy provides a structured framework for organising pronunciation errors into meaningful linguistic categories. Rather than analysing each error independently, taxonomy groups related pronunciation problems according to their phonetic or linguistic characteristics, enabling consistent classification across large datasets. Such structured classification resembles a lightweight ontology because relationships exist between broader error categories and their individual subtypes (Autayeu et al., 2010). This study adopts a taxonomy-based approach to classify pronunciation errors into phoneme substitution, stress and tone errors, consonant cluster errors and agglutination-related errors.

6.6 Summary and Research Gap

While neural TTS has improved for high-resource languages, indigenous African languages such as isiZulu continue to suffer from mispronunciation in navigation systems. Current research focuses on general-purpose TTS, leaving a gap in the systematic error analysis of isiZulu street names. This project addresses this by collecting a specialised corpus and establishing an automated evaluation framework to bridge the gap between commercial system output and native-speaker consensus.

7 Methodology

The study commenced with the collection of an initial dataset of verified isiZulu street names from the eThekwini Metropolitan Municipality, with particular emphasis on Umlazi region. To improve the scalability of the data collection process, a Python-based extraction pipeline was subsequently developed using OpenStreetMap (OSM) data. The automated extraction process produced more than 29 000 street name records from KwaZulu-Natal. The extracted data then underwent both automated and manual cleaning procedures to remove duplicate entries, numerical records, non-isiZulu street names and other invalid data. Following manual verification, a curated dataset of just over 1 000 verified isiZulu street names was produced for future integration into the NaVoice platform. This expanded dataset forms the foundation of the speech corpus being developed to support the pronunciation analysis and future development or adaptation of speech models for isiZulu street names.

Although the complete dataset contains more than 1 000 verified street names, the methodology presented in this study is initially validated using a representative subset of street names before being applied to the complete corpus. This incremental approach enables the proposed pronunciation analysis framework to be evaluated and refined while ensuring that it remains scalable for large-scale speech datasets.

7.1 Research Methodology

This study adopts a well-structured research methodology which will help with investigating pronunciation errors in isiZulu street names that are produced by existing navigation systems. The methodological decisions were guided by the Research Onion framework proposed by Saunders, Lewis & Thornhill (2019), and adapted by Mardiana (2020), which provides a systematic approach for selecting appropriate research philosophies, methodological choices and analytical procedures. Rather than focusing solely on software implementation, this framework ensures that the research process is theoretically grounded and aligned with the objectives of the study.

The study is underpinned by a pragmatic research philosophy, as it seeks to address a practical real-world problem through the development of an automated framework for analysing pronunciation errors in isiZulu street names. Pragmatism is particularly appropriate because the study combines established speech processing techniques with computational analysis to produce a practical solution that can contribute towards improving navigation tools for under-resourced African languages.

The study further follows a deductive research approach, whereby established speech processing techniques, such as the Mel-Frequency Cepstral Coefficients (MFCCs), Dynamic Time Warping (DTW), and acoustic feature analysis are applied to evaluate pronunciation performance on a new dataset of isiZulu street names. Existing theories and computational methods from speech processing literature will therefore guide the analysis and evaluation procedures adopted throughout the study.

Furthermore, a quantitative methodological choice is employed because pronunciation quality is assessed using objective numerical measurements rather that subjective human judgement (Abuhamda, Ismail & Bsharat, 2021). Acoustic similarity is quantified through Dynamic Time Warping, while complementary acoustic measurements, which includes pitch, duration and Root Mean Square (RMS) energy, would provide additional evidence for identifying different categories of pronunciation errors. These measurements will subsequently be combined into a composite Pronunciation Accuracy Score (PAS), allowing pronunciation quality to be evaluated consistently across multiple street names.

The research also adopts a design, implementation and evaluation strategy, in which an automated pronunciation analysis framework is developed and subsequently evaluated using speech recordings collected from existing navigation systems and native isiZulu speakers. The proposed framework is designed to automate pronunciation comparison, error classification and statistical analysis while remaining scalable for larger speech datasets that may be collected during future stages of the project.

Finally, the study will follow a cross-sectional time horizon, as the collected pronunciation recordings are analysed within a single period of investigation rather than being observed over an extended duration. The techniques and procedures implemented within this methodological framework include dataset development, audio pre-processing, acoustic feature extraction, pronunciation comparison, automatic error classification and statistical analysis of pronunciation patterns. These procedures collectively form the analytical framework developed for this research which are discussed in the following sections.

The overall research methodology adopted in this study is summarised in Figure 7.1. Instead of representing all the methodological options available within the Research Onion framework, the figure highlights the specific pathway followed in this research, beginning with the underlying research philosophy and then progress to the practical techniques and procedures implemented during the pronunciation analysis.

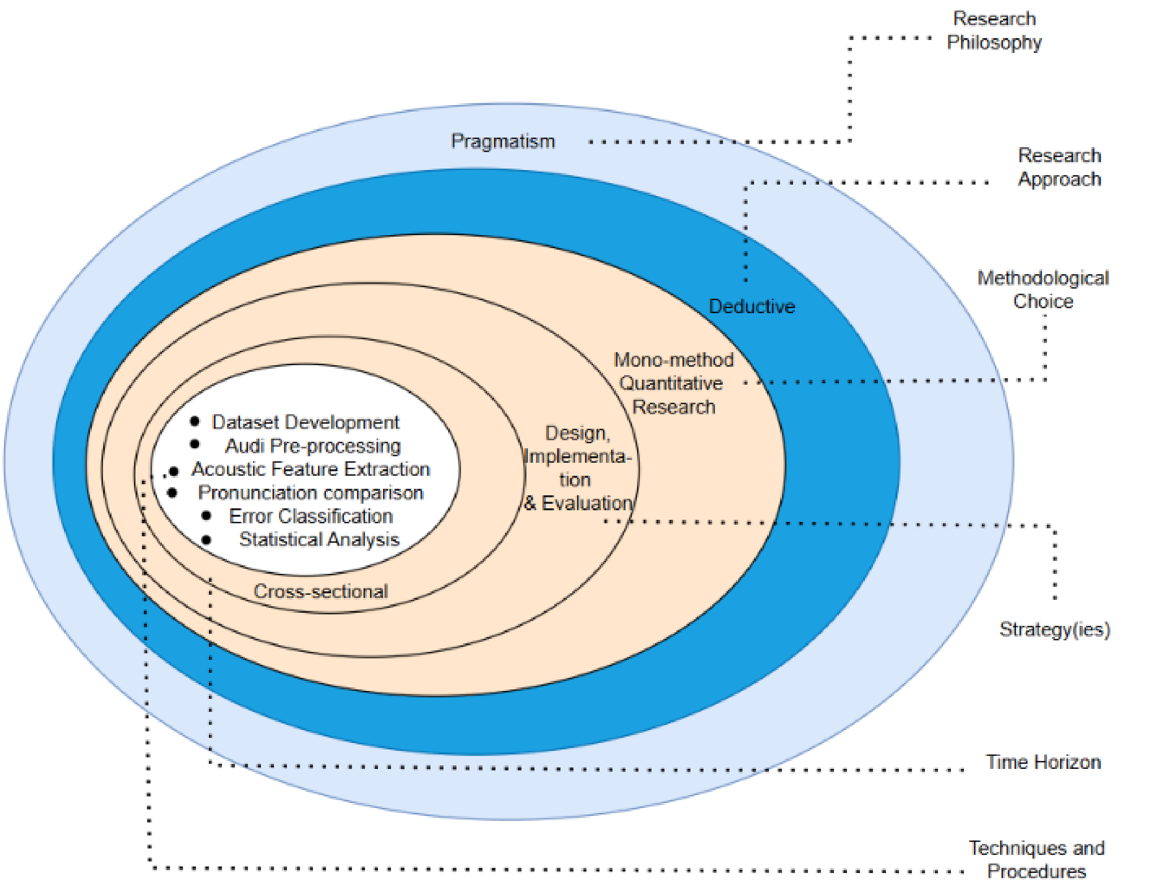

Figure 7.1: Research methodology adopted in this study, adapted from the Research Onion framework of Saunders et al. (2019) and Mardiana (2020).

Figure 7.1 illustrates the methodological decisions adopted throughout this study. These choices provide a theoretical foundation for the technical procedures presented in the sections that follow, ensuring that both the computational and the subsequent analysis are conducted within a coherent and academically rigorous research framework.

7.2 Research Design and Analytical Framework

This study adopted a quantitative experimental research design which allows the investigation of existing speech-based tools on how they pronounce isiZulu street names. Experimental research is appropriate where computational techniques are used to evaluate system performance under controlled conditions and objective measurements are obtained through repeatable analytical procedures (Creswell & Creswell, 2018). In context of speech technology, quantitative evaluation enables pronunciation differences between system-generated and human-produced speech to be measured using established acoustic metrics rather than relying solely on subjective human judgement (Jurafsky & Martin, 2026)

The study combines computational speech analysis with manual linguistic evaluation to identify, classify and analyse pronunciation errors that occur in navigation tools. The primary objective is not only to determine whether pronunciation errors occur, but also to understand the linguistic characteristics of these errors and evaluate their frequency and distribution across the collected dataset. This mixed analytical strategy is widely used in speech processing research, where objective acoustic measurements are complemented by expert linguistic interpretation to which in turn provides a more comprehensive evaluation of pronunciation quality (Taylor, 2009).

The research was conducted in successive stages. The first stage involved collecting and preparing a pronunciation dataset, this dataset consisted of verified isiZulu street names together with their corresponding pronunciations which were recorded by native isiZulu speakers and also system-generated pronunciations obtained from Apple Maps. The second stage then focused on developing an automated pronunciation analysis pipeline capable of pre-processing audio recordings, extracting acoustic speech features and comparing the system output with the corresponding ground truth recordings obtained from the NaVoice platform. The final stage involved analysing the resulting pronunciation differences, identifying recurring pronunciation patterns and constructing an error taxonomy that describes the observed errors.

This study combines qualitative language interpretation with objective acoustic measures instead of just depending entirely on subjective listening. Acoustic similarity between system pronunciations and native-speaker recordings is measured by Dynamic Time Warping (DTW), Mel-Frequency Cepstral Coefficients (MFCCs), pitch variation, speech duration and signal energy. These acoustic features have been widely used in speech processing because they capture complementary characteristics of human speech such as spectral information, temporal alignment and prosodic variation (Rabiner & Juang, 1993; Jurafsky & Martin, 2026).

A consensus ground truth approach was adopted because multiple native-speaker recordings for each street name were available, instead of relying on a single speaker as the pronunciation reference. Integrating multiple recordings reduces the influence of individual speaker variation and provides a more representative pronunciation reference for evaluation. Previous speech evaluation studies have shown that multiple reference pronunciations improve the robustness and reliability of pronunciation assessment by reducing speaker-specific bias (Taylor, 2009).

To improve efficiency and also ensure scalability, an automated analysis pipeline was developed in Python, The script automatically pre-processes speech recordings, extracts acoustic features, calculates pronunciation similarity metrics, classifies pronunciation deviations and generates summary reports, visualisations and statistical outputs for subsequent analysis. Although the script is capable of analysing the complete dataset automatically, detailed waveform, spectrogram and MFCC analyses are presented only for a representative sample of recordings, while the remaining street names are evaluated using the automatically generated quantitative metrics. This approach balances detailed qualitative interpretation with large-scale computational analysis and is consistent with experimental evaluation practices commonly adopted in speech technology research (Jurafsky & Martin, 2026).

The overall analytical framework adopted in this study is illustrated in Figure 7,2. The framework shows the sequence of stages followed during the research, beginning with the collection of verified isiZulu street names, then the ground truth pronunciations together with system-generated outputs. Followed by audio pre-processing and standardisation, acoustic feature extraction, pronunciation comparison and similarity measurement, automatic pronunciation error classification, manual validation of representative samples, and the construction of the final pronunciation error taxonomy. The framework concludes with quantitative evaluation, statistical analysis and visualisation and the findings to support the future development or adaptation of speech models for isiZulu street names.

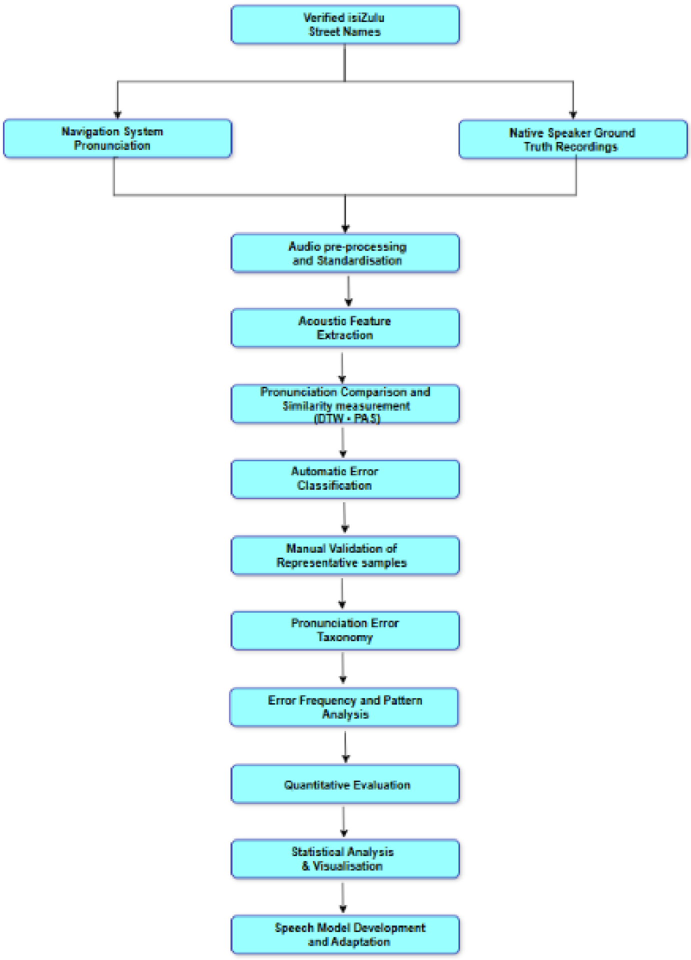

Figure 7.2: Overall research framework illustrating the pronunciation evaluation pipeline developed for analysing isiZulu street names from data collection through speech model development.

Figure 7.2 illustrates the complete research workflow developed for this study. The comparison results are then used to automatically identify pronunciation deviations, which are subsequently validated through manual linguistics inspection of representative samples and organised into a structured pronunciation error taxonomy. The classified errors are subjected to quantitative evaluation and statistical analysis to identify recuring pronunciation patterns and inform the future development or adaptation of speech models for isiZulu street names.

7.3 Dataset Description

The dataset used in this study consists of verified isiZulu street names collected from eThekwini Metropolitan Municipality and the Umlazi township in KwaZulu Natal, South Africa. The dataset forms the basis for evaluating the mispronunciation of isiZulu street names from existing navigation systems by comparing system generated pronunciations with corresponding ground truth recordings produced by native isiZulu speakers.

7.4 Data Collection

The initial phase of data collection focused on 150 verified isiZulu street names that were submitted to the NaVoice recording platform. Native speakers contributed multiple recordings for each street name, which guaranteed several ground truth pronunciations that captured natural variations in pronunciation between speakers, as highlighted by Taylor (2009), multiple recordings capture natural variations between speakers, resulting in a more representative pronunciation reference.

For the current stage of this study, Google Maps and the traditional GPS tools were selected as the navigation sys Apple Maps was selected as the navigation system under investigation. However, these tools required actually driving around and going to the destination to be able to record the systems pronunciation of street names. This was not efficient and effective hence; Apple Maps was selected because of it did not require driving around and also because it consistently produced audible pronunciation that could be recorded and converted into a format suitable for computational analysis.

Each street name in the dataset consists of three primary components: the textual street name, one system generated pronunciation, and one or more corresponding ground truth recordings. The use of multiple native speaker recordings enables the construction of a consensus ground truth, thereby reducing speaker specific variation and providing a more representative pronunciation reference for evaluating the system generated speech.

To support automated processing, each audio recording was standardised to WAV format with a sampling rate of 16 kHz prior to feature extraction. Speech processing studies commonly standardise recordings to a uniform sampling rate before feature extraction to ensure consistency during acoustic analysis (Pudasaini et al., 2025). The recordings were organised into a structured directory hierarchy in which each street name contains one folder for the system pronunciation and a corresponding folder containing all available native speaker recordings. This organisation enabled the automated analysis pipeline to process multiple street names without requiring manual intervention.

In addition to the audio recordings, metadata were maintained for each pronunciation sample, including the street name, language, geographic location, contributor identifier, recording format, sampling rate, recording device (where available), and the navigation system from which the pronunciation was obtained from. These metadata support reproducibility of the experiments and facilitates subsequent statistical analyses of pronunciation performance across different street names and recording conditions (Bird, Klein & Loper, 2009).

The metadata collected for each pronunciation sample are summarised in Table 7.1. These metadata provide the contextual information required for organising the pronunciation corpus, linking each system generated pronunciation with its corresponding ground truth recordings, and ensuring that the experiments can be reproduced.

Table 7.1: Metadata collected for each pronunciation sample.

Metadata Field	Description	Example
Street Name	Verified isiZulu street name	Mgaga Road
Language	Language represented	isiZulu
Geographic Location	Municipality of area	Umlazi
Province	Province of collection	KwaZulu Natal
Navigation System	Source of system pronunciation	Apple Maps
Speaker ID	Anonymous speaker identifier	Speaker001
Speaker Sex	Gender of contributor (where available)	Female
Approximate Age Group	Age range of contributor (where available)	35 – 44 years
System Audio	System generated pronunciation	Mgaga Road
Ground Truth Audio	Native speaker pronunciation	Speaker001.wav
Audio Format	Audio storage format	WAV
Sampling Rate	Recording sampling rate	16 kHz
Recording Device	Device used for recording (if known)	Smart Phone
Dialect or Home Region	Primary isiZulu Dialect or Region spoken by contributor (where Available)	eThekwini

The pronunciation dataset shown in Table 7.1. will continue to expand as additional street names and native speaker recordings are collected, the analysis framework developed in this study has been designed to operate automatically on multiple street names.

7.5 Dataset Selection for Experimental Evaluation

Although more than 100 isiZulu street names were collected and prepared for analysis, only 20 representative street names were selected for detailed experimental evaluation in this study. This subset was chosen to provide a manageable yet linguistically diverse sample containing variations in syllable structure, consonant clusters, click consonants, agglutinative morphology, and vowel combinations. Analysing a representative subset enabled detailed qualitative and quantitative error analysis while remaining within the scope and time constraints of this research project. Furthermore, the selected street names represented a range of pronunciation challenges commonly encountered in isiZulu place names, allowing the proposed evaluation framework to assess multiple categories of pronunciation errors. Although only 20 street names are discussed in detail within this study, the developed analysis pipeline is fully automated and can be applied to the remaining collected street names in future work using the same methodology.

7.6 Audio Pre-processing and standardisation

Speech recordings obtained from both the native speaker contributors and the navigation system were not immediately suitable for acoustic comparison because they exhibited variations in recording quality, background silence, signal amplitude and speaking duration. Such inconsistences are common in speech corpora collected multiple speakers and recording environments and can significantly influence the extraction of acoustic features if they are not addressed before analysis (Jurafsky & Martin, 2026; Kalahroodi, Faili & Shakery, 2026).

To improve the consistency of the pronunciation analysis, an automated audio pre-processing pipeline was developed in Python using the Librosa speech processing library. The pipeline was designed to standardise all recordings prior to feature extraction and pronunciation comparison. Automating the pre-processing stage also ensures that identical processing procedures were applied to every pronunciation sample, thereby improving the reproducibility and reliability of the experiments (Reddy et al., 2026).

The pre-processing pipeline consisted of four sequential stages. First, all recordings were converted into a uniform waveform audio format (WAV) with a sampling rate of 16 kHz. Standardising the sampling rate ensured that acoustic features extracted from different recordings were directly comparable and eliminated inconsistencies caused by different recording devices or encoding formats (Mcfee et al., 2015).

Secondly, leading and trailing silence was automatically removed using Librosa's silence trimming algorithm. The removal of silent segments reduced the influence of recording delays that naturally occurred before speakers began pronouncing the street names or after they had finished speaking. Without this step, speech duration measurements and acoustic similarity scores could become artificially distorted because Dynamic Time Warping would attempt to align silence rather than the spoken pronunciation itself.

Thirdly, each recording was normalised to a common amplitude level. Volume normalisation reduced differences caused by recording distance, microphone sensitivity and speaker loudness, ensuring that subsequent acoustic analyses focused on pronunciation characteristics rather than signal intensity.

Finally, the processed recordings were inspected visually through waveform representations to verify that silence had been removed while preserving the complete pronunciation of the street name. Visual inspection formed an important quality assurance step before acoustic feature extraction was performed.

The effect of the pre-processing pipeline is illustrated in Figure 7.3 and Figure 7.4. Figure 7.3 presents the original waveform prior to pre-processing, where noticeable periods of silence are visible before and after the spoken pronunciation. Figure 7.4 shows the same recording after automatic silence removal and signal normalisation. The reduction of non-speech regions produces a cleaner speech signal, thereby improving the reliability of the subsequent pronunciation comparison.

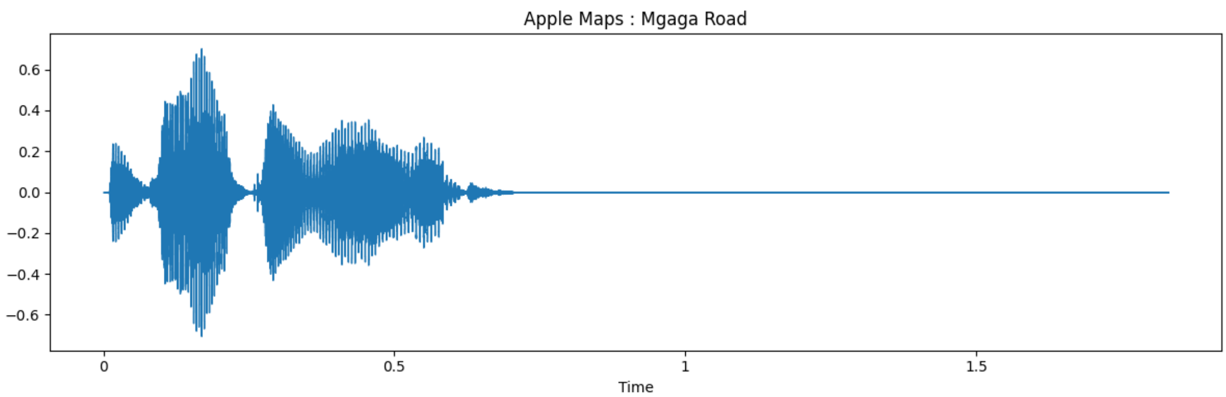

Figure 7.3: Original waveform of the Apple Maps pronunciation before audio pre-processing.

As shown in Figure 7.3, the original recording contains leading and trailing silence surrounding the spoken pronunciation. These silent regions do not contribute meaningful pronunciation information and may influence duration measurements and temporal alignment during pronunciation comparison. Consequently, automatic pre-processing was applied before feature extraction.

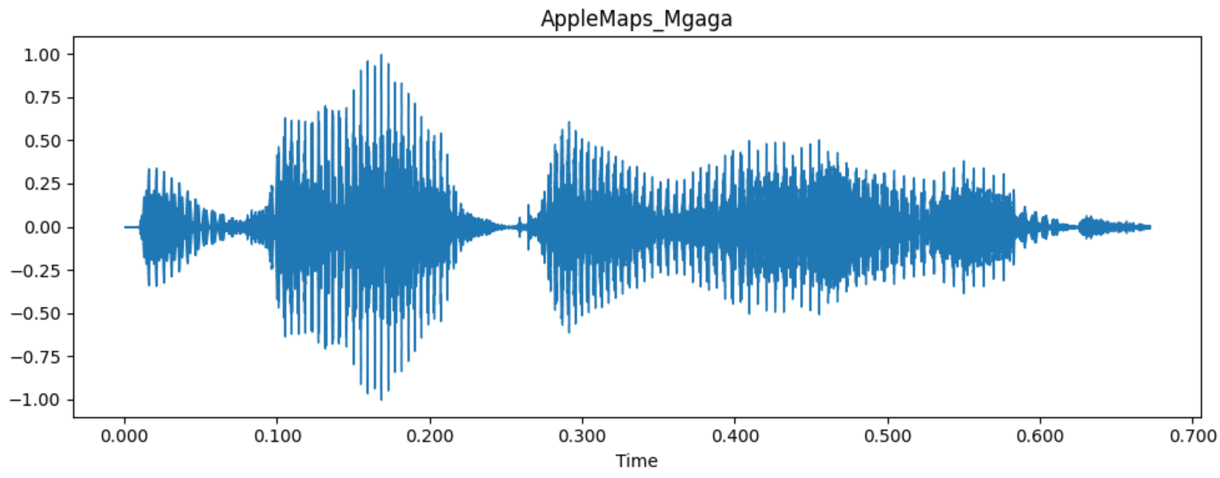

Figure 7.4: Waveform after automatic silence trimming and amplitude normalisation.

Figure 7.4 demonstrates the outcome of the pre-processing stage. The non-speech regions have been removed while preserving the pronunciation itself. This results in a cleaner acoustic signal that provides a more reliable basis for extracting pronunciation features such as MFCCs, pitch, duration and energy.

7.7 Acoustic Feature Extraction

Instead of comparing the raw audio data waveforms directly, feature extraction transforms the speech signal into numerical representations that capture important characteristics of pronunciation while reducing the influence of recording conditions and background variability. Acoustic feature extraction is a fundamental stage in speech processing because it enables objective comparison between different speech recordings (Dmitriev et al., 2018). This study extracted four complementary acoustic features from every pronunciation recording, namely Mel-Frequency Cepstral Coefficients (MFCCs), pitch, duration and collectively provides a comprehensive representation of pronunciation characteristics.

The extraction process was implemented using the Librosa Python library, allowing every system generated pronunciation and corresponding ground truth recording to be processed automatically using identical parameters. Standardising the feature extraction process ensured that all recordings were analysed consistently throughout the pronunciation comparison pipeline.

7.7.1 Mel-Frequency Cepstral Coefficients (MFCCs)

Mel-Frequency Cepstral Coefficients (MFCCs) are among the most widely used acoustic features in speech recognition and speech synthesis because they provide a compact representation of the spectral characteristics of human speech. Unlike raw waveforms, MFCCs approximate the way the human auditory system perceives sound by representing speech on the perceptually motivated Mel frequency (Davis & Mermelstein, 1980).

The computation of MFCCs consists of several signal-processing stages including framing, windowing, Fast Fourier Transform (FFT), Mel filter bank analysis, logarithmic compression and Discrete Cosine Transform (DCT). The resulting cepstral coefficients are given by:

$$\mathrm{C_n} = \sum_{k-1}^{k} log(S_k) cos\left[n\left(\mathrm{k} - \frac{1}{2}\right)\frac{\pi}{K}\right]$$

Where:

- $\mathrm{C_n}$ is the n[th] MFCC coefficient
- S_k is the Mel filter bank energy
- K is the total number of Mel filters
- n = 1, ...,13

In this study, after following common practice in speech processing, thirteen Mel-Frequency Cepstral Coefficients (MFCCs) were extracted from each recording after pre-processing because they provide a compact representation of the speech spectrum while discarding unnecessary acoustic variations (Davis & Mermelstein, 1980; Jurafsky & Martin, 2026).

The MFCC representation extracted from the processed Apple Maps pronunciation is illustrated in Figure 7.5. The figure demonstrates how speech signal is transformed from a waveform into a compact spectral representation suitable for computational comparison.

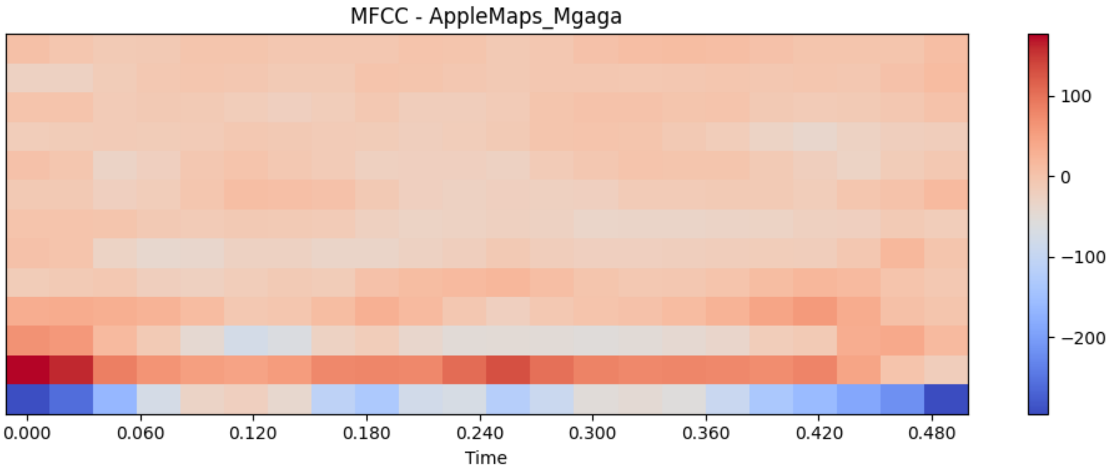

Figure 7.5: MFCC representation of the pre-processed Apple Maps pronunciation.

The x-axis represents time, while the vertical y-axis represents the MFCC coefficient number. The colour intensity indicates the magnitude of each coefficient, with warmer colours representing stronger spectral energy. Changes in colour patterns correspond to changes in pronunciations using Dynamic Time Warping.

7.7.2 Pitch Extraction

Pitch represents the perceived fundamental frequency of speech and is associated with vocal fold vibration during pronunciation. Although isiZulu is not a tonal language in the same sense as many Asian languages, variations in pitch remain important because they influence natural speech prosody, emphasis and pronunciation quality (Taylor, 2009).

The fundamental frequency (F0) was estimated using the YIN algorithm implemented in the Librosa library. The search range was restricted to 75-400 Hz, corresponding to the expected frequency range of human speech. Moreover, the differences between the average pitch of the navigation system and the native speaker recordings were subsequently calculated to identify possible stress or intonation differences.

Pitch was therefore treated as a complementary pronunciation feature rather than the primary comparison metric. Large pitch deviations may indicate that the navigation system applies unnatural emphasis during pronunciation even when the sequence of phonemes remain relatively accurate.

7.7.3 Duration Extraction

Speech duration measures the length of time required to pronounce an entire street name. Differences in pronunciation duration may include slower articulation, elongated vowels, omitted speech segments or difficulties associated with agglutinative word structures.

Following silence removal, the duration off every recording was computed automatically. Duration differences between the system generated pronunciation and ground truth recordings were subsequently included as one of the components of the Pronunciation Accuracy Score (PAS). Because the pre-processing stage removed leading and trailing silence, the measured durations primarily reflected actual pronunciation rather than recording delays.

7.7.4 Energy Extraction

Speech energy describes the intensity of the speech signal throughout the pronunciation. Even though the overall loudness is not a direct indicator of pronunciation quality, abrupt changes in energy may indicate unstable speech production, clipped pronunciations or recording inconsistencies.

The Root Mean Square (RMS) energy was then extracted from every recording to provide an additional measure of signal consistency. Energy differences were not used independently to classify pronunciation errors but were incorporated into the overall pronunciation analysis to improve the robustness of the comparison process.

The acoustic features extracted from each pronunciation recording and their respective roles in the analysis are summarised in Table 7.2.

Table 7.2: Acoustic features extracted and their role in the pronunciation analysis framework.

Feature	Description	Role in Pronunciation Analysis
Mel-Frequency Cepstral Coefficients (MFCCs)	Compact representation of the spectral characteristics of speech based on human auditory perception.	Used as the primary acoustic representation for Dynamic Time Warping (DTW) to measure pronunciation similarity between system-generated and ground truth recording
Pitch (Fundamental Frequency, F0)	Represents the average fundamental frequency of the speech signal.	Used to identify differences in stress placement and intonation between pronunciations.
Duration	Measures the temporal length of each pronunciation.	Used to detect timing differences that may indicate pronunciation variation or agglutination-related effects.
Root Mean Square (RMS) Energy	Measures the average energy or intensity of the speech signal.	Used to evaluate recording consistency and contributes to the computation of the Pronunciation Accuracy Score (PAS).

As shown in Table 7.2, no single acoustic feature is sufficient to describe pronunciation quality. But the extracted features complement one another by representing different characteristics of speech production. This multi-feature approach provides a more comprehensive basis for pronunciation comparison than relying on a single measurement alone.

7.8 Pronunciation Comparison Framework

Following acoustic feature extraction, the system generated pronunciations were quantitatively compared with the corresponding native speaker recordings, in place of relying on subjective listening alone, an automated pronunciation comparison framework was developed to evaluate pronunciation similarities using multiple complementary acoustic measures. The framework was implemented in Python and formed the core analytical component of this study.

The comparison framework was designed to accommodate multiple ground truth recordings for every street name. Since individual speakers naturally differ in speaking rate, pitch, pronunciation style and recording conditions, comparing the navigation system against a single speaker could introduce speaker specific bias (Patel et al., 2025). Instead, the proposed framework compares the system pronunciation with every available native speaker recording before aggregating the results to obtain a representative pronunciation assessment. This approach improves the reliability of the evaluation while reducing the influence of individual speaker variation.

The pronunciation comparison process consisted of four principal stages, the Mel-Frequency Cepstral Coefficients (MFCCs) extracted from the system pronunciation and the corresponding ground recording were aligned using Dynamic Time Warping (DTW). Subsequent to that, energy differences were calculated. These measures were then combined into a composite Pronunciation Accuracy Score (PAS). Finally, the computed metrics were used to identify potential pronunciation errors and assign each pronunciation to one or more predefined error categories.

The overall pronunciation comparison process implemented in this study is illustrated in Figure 7.6.

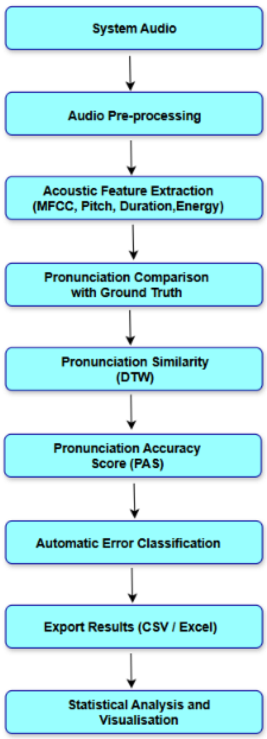

Figure 7.6: Pronunciation comparison framework developed for this study.

The framework illustrated in Figure 7.6 automatically compares each navigation system pronunciation with the corresponding native speaker ground truth recordings, computes multiple metrics and assigns pronunciation error categories, then exports results for subsequent statistical analysis and visualisation.

7.8.1 Dynamic Time Warping

Dynamic Time Warping (DTW) was employed as the primary pronunciation similarity measure because different speakers seldom pronounce the same word at identical speaking rates. Conventional distance measures such as Euclidean distance , require both signals to have equal duration and identical temporal alignment. In contrast, DTW aligns speech sequences by allowing non-linear stretching and compression along the time axis, which in turn produces a more robust similarity measure for speech signals (Jiang & Chen, 2023).

In this study DTW was applied to MFCC representations extracted from the system pronunciation and the native speaker recordings/ The accumulated DTW cost was subsequently normalised by the length of the optimal warping path to ensure that recordings of different durations remained directly comparable.

The accumulated DTW distance is computed recursively as:

$$DTW\ (i, j) = d\ (i, j) + min \left\{ \begin{matrix} DTW(i - 1, j) \\ DTW(i, j - 1) \\ DTW(i - 1, j - 1) \end{matrix} \right\}$$

Where:

d (i, j) represents the local distance between MFCC feature vectors;

DTW (i, j) denotes the cumulative alignment cost;

i and ***j*** represent the feature indices of the two speech sequences.

The final DTW score was then normalised using

$$DTW_{norm} = \frac{DTWtotal}{L}$$

Where:

DTW_{total} is the accumulated DTW distance;

L is the number of points contained within the optimal warping path

Normalisation prevents longer recordings from automatically producing larger DTW values and thus improves comparability across different speakers.

7.8.2 Pronunciation Accuracy Score (PAS)

Although DTW provides an effective measure of pronunciation similarity, pronunciation quality cannot be fully described by spectral similarity alone. Consequently, a composite Pronunciation Accuracy Score (PAS) was developed to combine several combine several complementary acoustic measures into a single performance indicator.

The PAS integrates four pronunciation components:

i. Dynamic Time Warping similarity.
ii. Pitch similarity.
iii. Duration similarity.
iv. Energy similarity.

Before calculating the composite Pronunciation Accuracy Score, each acoustic measure was normalised to a common similarity scale ranging from 0 to 100 to ensure that no single metric dominated the final score because of differences in numerical magnitude.

The composite score is calculated as:

$$\boldsymbol{PAS = 0.5D + 0.2T + 0.2E + 0.1P}$$

Where:

D represents the normalised DTW similarity;

T represents duration similarity;

E represents energy similarity;

P represents pitch similarity.

The weighting scheme was intentionally designed to reflect the relative importance of each acoustic feature in pronunciation assessment. DTW, computed from the MFCC representations, was assigned the highest weighting (50%) because it captures the phonetic content and overall acoustic similarity between the system pronunciation and the native speaker recordings. Duration and energy were each assigned a weighting of 20% because they provide complementary information regarding pronunciation timing and signal consistency, both of which contribute to the pronunciation quality. Pitch was assigned 10%, as natural variations in speaker intonation are expected even among native speakers, making pitch a less reliable indicator of pronunciation accuracy when considered in isolation. Consequently, the proposed weighting scheme prioritises phonetic similarity while still incorporating prosodic and temporal characteristics into the overall PAS. The resulting PAS ranges between 0 and 100, with higher values indicating greater similarity between the navigation system pronunciation and the corresponding native speaker pronunciation.

Following acoustic feature extraction, several quantitative pronunciation metrics were computed to evaluate the similarity between navigation system pronunciations and the corresponding native speaker ground truth recordings. Each metric captures a different aspect of pronunciation performance and collectively contributes to a comprehensive assessment of pronunciation accuracy. The pronunciation metrics calculated by the proposed framework are summarised in Table 7.3.

Table 7.3: Pronunciation metrics calculated by the proposed framework

Metric	Purpose	Contribution to pronunciation Analysis
Dynamic Time Warping (DTW)	Measures the similarity between the MFCC feature sequences of the navigation system pronunciation and the native-speaker ground truth.	Primary metric for quantifying phonetic pronunciation similarity.
Pitch Difference (F0)	Measures the difference in average fundamental frequency between recordings	Identifies stress placement and intonation differences between pronunciations
Duration Difference	Measures the difference in pronunciation length between recordings.	Detects timing variations that may indicate agglutination-related or rhythm-related pronunciation differences.
Root Mean Square (RMS) Energy Difference	Measures differences in the average signal energy of the recordings.	Evaluates recording consistency and contributes to the composite pronunciation score.
Pronunciation Accuracy Score (PAS)	Combines DTW, pitch, duration and RMS energy into a single weighted score.	Provides the overall quantitative measure of pronunciation accuracy used throughout this study

As shown in Table 7.3, no single metric is sufficient to characterise pronunciation accuracy. DTW provides the primary measure of phonetic similarity, while pitch, duration and RMS energy quantify complementary prosodic and temporal characteristics of speech.

7.9 Pronunciation Error Classification and Taxonomy

The objective of this stage was not only to determine whether a pronunciation differed from native speaker reference, but also to identify the most likely linguistic source of the error. This approach provides a more informative evaluation that solely relying on numerical scores because it allows pronunciation failures to be analysed according to their linguistic characteristics. The overall pronunciation error classification process is illustrated in Figure 7.7.

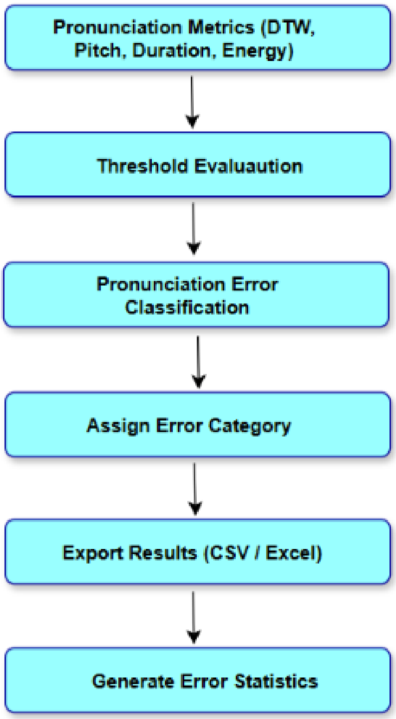

Figure 7.7: Rule-based pronunciation error classification framework developed for this study.

Figure 7.7 illustrates the rule-based pronunciation error classification framework implemented in this study. The framework receives the calculated pronunciation metrics as the input and then applies predefined threshold-based decision rules to determine the most likely pronunciation error category. Once the error category has been assigned, the results are exported for subsequent statistical analysis and visualisation. Automating the classification process ensures that identical decision criteria are applied consistently across all analysed street names, thereby improving the repeatability and objectivity of the pronunciation evaluation.

7.9.1 Dynamic Time Warping- Based Pronunciation Classification

Dynamic Time Warping (DTW is the principal pronunciation similarity measure adopted in this study. DTW is a well-established algorithm for comparing two temporal sequences that may differ in speaking rate or duration by finding an optimal alignment between them (Jiang & Chen, 2023). In speech processing, pronunciation recordings of the same word are rarely identical because speakers naturally vary in their speaking speed, rhythm and articulation. Jiang and Chen (2023) further alluded that, direct frame-by-frame comparison of speech signals often produce misleading similarity measurements and that DTW overcomes this limitation by allowing non-linear alignment of corresponding speech segments before calculating the overall pronunciation distance.

The DTW distance provides the primary quantitative indication of pronunciation similarity. Lower DTW values indicate that the system-generated pronunciation closely matches the native-speaker pronunciation, whereas larger DTW values suggest greater pronunciation deviations. However, DTW alone cannot determine the linguistic nature of the pronunciation error. For example, a high DTW score may result from incorrect phoneme production, misplaced stress, timing differences or a combination of several pronunciation deviations. Consequently, additional acoustic measurements, including pitch difference, duration difference and RMS energy difference, are incorporated into the analysis framework to provide complementary evidence during pronunciation error classification.

Although Dynamic Time Warping provides the primary measure of pronunciation similarity, the interpretation of pronunciation deviations requires linguistic error categories that describe the underlying characteristics of the observed errors. The pronunciation error categories investigated in this study are summarised in Table 7.4.

Table 7.4: Summarises the pronunciation error categories investigated in this study.

Error Category	Description	Primary Detection Metric(s)	Interpretation Threshold
Phoneme Substitution	Incorrect pronunciation or substitution of one or more speech sounds relative to the native-speaker pronunciation	Dynamic Time Warping (DTW)	DTW > 170
Stress / Tone Error	Incorrect placement of stress or deviations in pitch and intonation patterns	Pitch Difference	Pitch Difference > 80 Hz
Consonant Cluster Error	Mispronunciation involving consonant sequences or consonant combinations commonly found in isiZulu street names	DTW + Spectrogram Inspection	DTW > 170 plus spectrogram evidence
Agglutination related Error	Pronunciation errors associated with isiZulu agglutinative morphology, including incorrect timing or merging of morphemic units.	Duration Difference + DTW	Duration Difference > 0.8 s
Multiple Errors	Simultaneous occurrence of two or more pronunciation error categories within the same pronunciation.	Combination of Multiple Metrics	Multiple thresholds exceeded

As shown in Table 7.4, each pronunciation error category is associated with one or more acoustic measures. While DTW provides an overall measure of pronunciation similarity, complementary measures such as pitch and duration assist in distinguishing between different types of pronunciation errors.

Pronunciation errors identified during manual analysis were classified using a structured error taxonomy consisting of substitution, deletion and insertion errors. Substitution errors were further categorised according to the affected linguistic feature, including vowel substitution, consonant substitution, click consonant substitution, consonant cluster simplification and agglutination-related pronunciation errors.

7.9.2 Pronunciation Error Taxonomy

The pronunciation error categories presented in Table 7.4 can be organised into a hierarchical taxonomy based on the underlying linguistic characteristics of the observed pronunciation deviations. This taxonomy provides a structured representation of pronunciation errors identified during the automatic evaluation process and forms the conceptual foundation of the proposed pronunciation classification framework. The resulting taxonomy is illustrated in Figure 7.8.

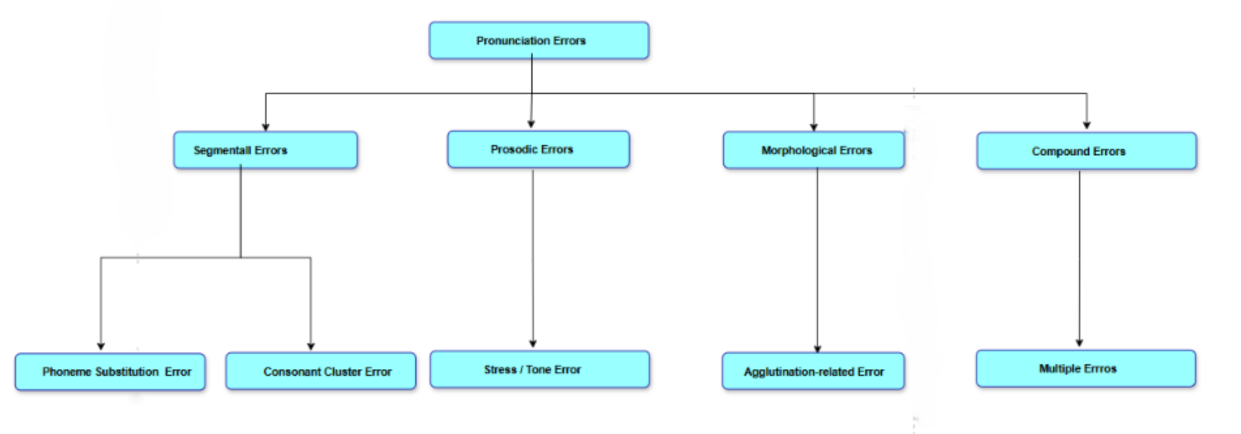

Figure 7.8: Hierarchical taxonomy of pronunciation error categories developed for the automatic classification of isiZulu street name pronunciations.

Figure 7.8 illustrates the pronunciation error taxonomy developed in this study. The taxonomy organises pronunciation deviations into four primary linguistic categories, namely segmental, prosodic, morphological and compound errors. Segmental errors include phoneme substitution and consonant cluster errors, while prosodic errors describe deviations in stress placement or intonation. Morphological errors capture pronunciation deviations associated with the agglutinative structure of isiZulu, whereas compound errors represent pronunciations exhibiting multiple error categories simultaneously. This hierarchical organisation facilitates consistent interpretation of automatically detected pronunciation errors and provides a foundation for the quantitative analyses presented in the subsequent chapter.

7.10 Frequency and Pattern Analysis

After individual pronunciation errors had been identified and classified, the resulting metrics were aggregated to examine the overall performance of the navigation system across the dataset. Rather than evaluating each street name independently, the analysis focused on identifying recurring pronunciation patterns and determining the types of pronunciation errors that occurred frequently. This approach enabled a broader assessment of the strengths and limitations of the speech synthesis system when pronouncing isiZulu street names.

The analysis combined quantitative performance measures with visualisation techniques to facilitate the interpretation of pronunciation errors. The computed pronunciation metrics and error classifications were automatically exported to Microsoft Excel and comma-separated value (CSV) files for further statistical analysis. In addition, following the automatic classification of pronunciation errors, the resulting error categories and pronunciation metrics are aggregated to identify recurring error patterns across the dataset. The analytical workflow used to summarise pronunciation performance, generate statistical visualisations and support the interpretation of results is presented in Figure 7.9.

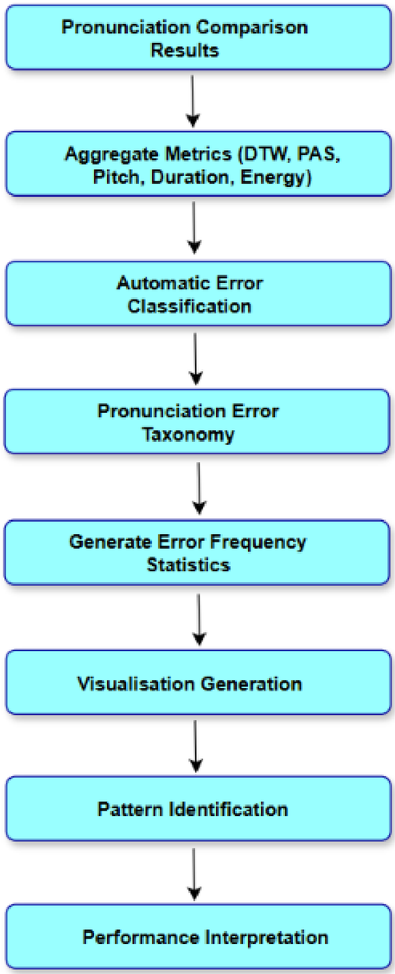

Figure 7.9: Workflow used to analyse pronunciation frequency and error patterns across the dataset.

Figure 7.9 illustrates the workflow adopted for analysing pronunciation frequency and error patterns. Following pronunciation comparison and automatic error classification, the extracted metrics are aggregated to produce descriptive statistics, including error frequencies, pronunciation accuracy distributions and relationships between acoustic features. These statistical outputs provide the basis for identifying recurring pronunciation patterns and evaluating the performance of existing navigation systems across multiple isiZulu street names.

7.10.1 Statistical Analysis

The pronunciation comparison framework computes several descriptive statistics to summarise system performance across all analysed street names. These statistics include the mean, minimum, maximum and standard deviation of the Pronunciation Accuracy Score (PAS), Dynamic Time Warping (DTW), pitch difference, duration difference and energy difference. These summary statistics provide an overall indication of pronunciation consistency and enable comparisons between different pronunciation metrics.

In addition to numerical summaries, the framework automatically ranks street names according to their average PAS values. Streets with lower PAS values indicate poorer pronunciation agreement between the navigation system and the native-speaker recordings, thereby allowing problematic pronunciations to be identified for further qualitative analysis. Ranking the results also assists in selecting representative examples for detailed discussion without requiring manual inspection of every pronunciation.

Table 7.5: summarises the quantitative measures generated by the pronunciation comparison framework during the frequency and pattern analysis stage.

Measure	Purpose
Mean PAS	Overall pronunciation accuracy
Mean DTW	Average acoustic similarity
Mean Pitch Difference	Overall stress/tone variation
Mean Duration Difference	Average pronunciation timing variation
Mean Energy Difference	Average recording energy variation
Standard Deviation	Pronunciation consistency
Worst-performing Streets	Identification of problematic pronunciations
Best-performing Streets	Identification of highly accurate pronunciations

As shown in Table 7.5, both individual pronunciation measures and summary statistics were used to evaluate pronunciation performance. These descriptive statistics provide an overall assessment of pronunciation quality while facilitating comparisons between different street names and pronunciation error categories.

7.10.2 Visual Analysis

To complement the quantitative evaluation, the framework automatically generates several graphical visualisations that support the interpretation of pronunciation performance and pronunciation error patterns. Visual representations simplify the identification of trends that may not be immediately apparent from numerical tables alone and thus provides additional evidence during the analysis process.

The primary visualisations generated by the framework include:

- Error category frequency bar charts illustrating the distribution of pronunciation error types.
- Histograms showing the distribution of Pronunciation Accuracy Scores across the dataset.

- Scatter plots illustrating the relationship between Dynamic Time Warping and pitch difference.
- Bar charts ranking the lowest-performing street names according to their average PAS.
- Heatmaps summarising pronunciation similarity across all analysed street names and native-speaker recordings.

Rather than analysing every available pronunciation individually, approximately ten representative street names are selected for detailed visual discussion, while the remaining pronunciations are evaluated automatically using the quantitative measures produced by the framework. This sampling strategy is consistent with accepted research practice for large speech datasets and enables detailed interpretation without compromising scalability.

7.10.3 Acoustic Feature Extraction Parameters

To ensure methodological consistency and experimental reproducibility, all audio recordings were processed using a fixed set of pre-processing and feature extraction parameters throughout the study. These parameters were selected based on commonly adopted speech processing practices and remained unchanged for all pronunciation comparisons. Table 7.6 summarises the acoustic feature extraction and analysis parameters implemented within the proposed evaluation framework.

Table 7.6: Acoustic Feature Extraction and Analysis Parameters.

Parameter	Value	Purpose
Sampling Rate	16kHz	Standardises all recordings
Audio Format	WAV	Lossless speech processing
Silence Trimming	`librosa.effects.trim()`	Removes leading and trailing silence
Normalisation	Peak amplitude = 1.0	Removes loudness differences
MFCC Coefficients	13	Spectral representation of speech
FFT window (n_fft)	2048	Frequency-domain analysis
Hop length	512 Samples	Frame overlap for feature extraction
DTW distance	Euclidean	Measures pronunciation similarity
Pitch algorithm	`librosa.yin()`	Estimates fundamental frequency (F0)
Energy	RMS Energy	Measures speech intensity
Duration	Seconds	Measures total speech length

The parameter configuration presented in Table 7.6 was applied consistently across all recordings throughout the study. Standardising the pre-processing and feature extraction settings ensured that differences observed between Apple Maps pronunciations and native-speaker recordings reflected pronunciation characteristics rather than inconsistencies introduced during data processing. Consequently, these parameters provided a reliable and reproducible foundation for the objective pronunciation comparison, error classification, and statistical analyses presented in the preliminary results section.

7.11 Experimental Environment

All experiments, including audio pre-processing, feature extraction, pronunciation comparison, statistical analysis, and visualisation, were conducted using the computing environment described in Table 7.7. Reporting the experimental environment enhances methodological transparency and facilitates reproducibility of the proposed pronunciation evaluation framework.

Table 7.7: Experimental Computing Specifications.

Component	Specification
Laptop Model	Acer Aspire Go 15
Operating System	Window 11 (WSL2: Ubuntu 22.04)
Processor	13th Gen Intel® Core ™ i5 – 1334U
RAM	16 GB DDR5
GPU	Intel® UHD Graphics 16GB
Python Version	3.11.15
IDE	Visual Studio Code
Environment	Anaconda
Main Libraries	librosa, NumPy, Pandas, SciPy, Matplotlib, OpenPyXL
Storage	512 SSD(NVMe) External 2TB SSD(Toshiba NTFS)

The above computing environment provided sufficient computational resources to perform feature extraction, Dynamic Time Warping analysis, statistical evaluation, and visualisation without hardware-related limitations. All experiments were executed using identical software versions throughout the study to ensure consistency across analyses.

The complete Python implementation, experimental scripts, generated figures, and supporting files developed during this study are maintained in a GitHub repository to facilitate transparency and reproducibility. The repository URL is provided in Appendix D.

8 Preliminary Results

8.1 Overall Pronunciation Performance

The preliminary analysis was conducted to evaluate the pronunciation quality of Apple Maps for isiZulu street names using objective speech comparison metrics. Pronunciation performance was assessed using the Pronunciation Accuracy Score (PAS), Dynamic Time Warping (DTW), pitch, duration, and energy measurements extracted from the Apple Maps pronunciations and the corresponding native speaker recordings. These metrics collectively provide an initial indication of how closely the synthesised pronunciations resemble native isiZulu speech.

Table 8.1: Summary of Experimental Results

Metric	Result
Total street names analysed	20
Total native speaker recordings analysed	113
Mean Pronunciation Accuracy Score (PAS)	46.2%
Highest PAS	66.7%
Lowest PAS	32.4%
Mean DTW Distance	20871.79
Mean Normalised DTW	221.954
Mean Similarity	0.005
Mean Apple Maps Duration (s)	0.98
Mean Native Duration (s)	2.96
Mean Apple Maps Pitch (Hz)	185.97
Mean Native Pitch (Hz)	157.50
Mean Apple Maps Energy	0.1849
Mean Native Energy	0.0480

The summary presented in Table 8.1 provides an overview of the experimental dataset and the average pronunciation performance across all analysed street names. A total of 33 isiZulu street names were evaluated against 165 native speaker recordings**,** providing multiple pronunciation comparisons for each street name. The overall mean Pronunciation Accuracy Score (PAS) indicates that the synthesised pronunciations achieved only a moderate level of agreement with native pronunciations. Similarly, the average DTW distance and similarity scores suggest measurable acoustic differences between Apple Maps pronunciations and the corresponding native speaker recordings.

These findings provide quantitative evidence that even though Apple Maps is capable of producing recognisable pronunciations for several isiZulu street names, there is a considerable pronunciation variation that still remains across the dataset. Similar challenges have been reported in recent studies on speech technologies for low-resource African languages, where limited linguistic resources and insufficient language-specific training data negatively affect pronunciation quality and speech naturalness (Adebara & Abdul-Mageed, 2022; Bhushan *et al.*, 2025).

The overall distribution of Pronunciation Accuracy Scores is illustrated in Figure 8.1**.**

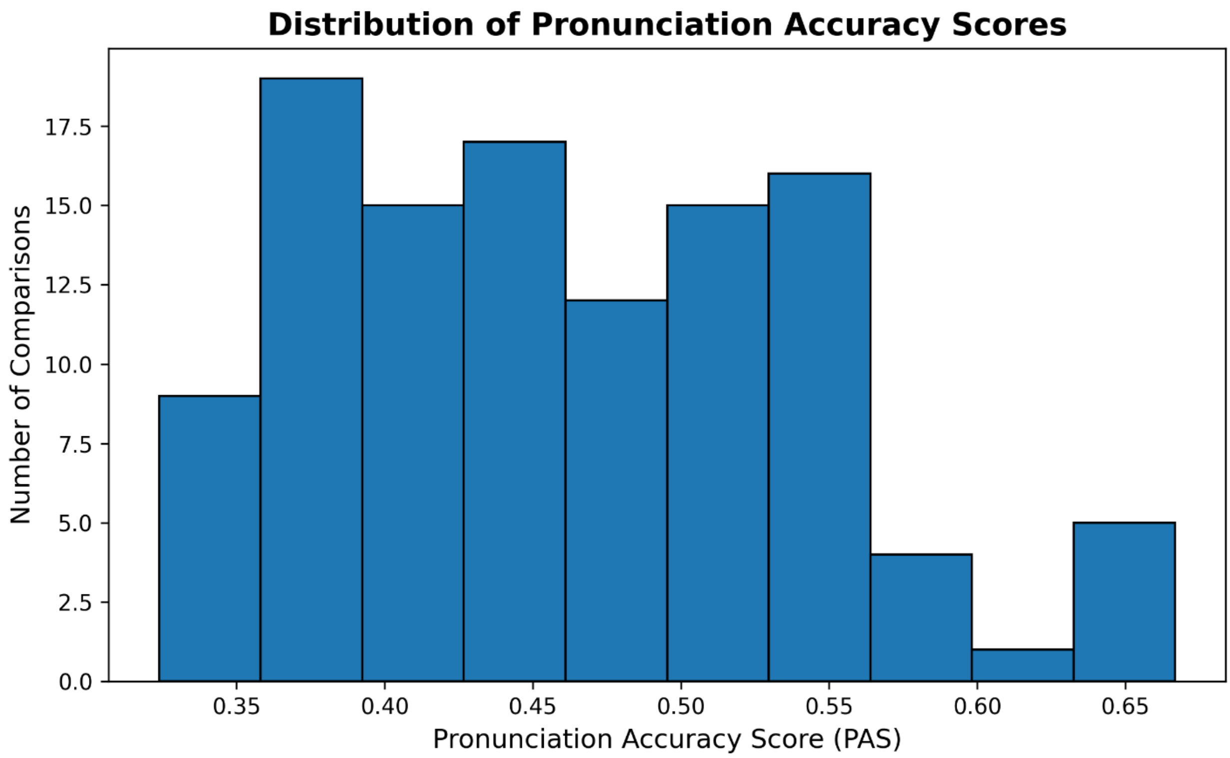

Figure 8.1: Distribution of Pronunciation Accuracy Scores (PAS).

Figure 8.1 illustrates the distribution of PAS values obtained across all pronunciation comparisons. The distribution demonstrates that pronunciation accuracy varies considerably among the analysed street names, with most scores concentrated within the middle range rather than near perfect accuracy. This indicates that Apple Maps does not consistently produce pronunciations that closely resemble native isiZulu speech.

The absence of a strong concentration of high PAS values further suggests that pronunciation performance remains inconsistent across different lexical items. Previous research has shown that pronunciation quality in under-resourced languages is strongly influenced by the availability of high-quality speech corpora and language-specific phonetic modelling, both of which remain limited for many African languages (Besacier et al., 2014; Adebara et al., 2023).

To complement the PAS analysis, Dynamic Time Warping (DTW) distances were examined to quantify the acoustic similarity between Apple Maps pronunciations and the corresponding native speaker recordings.

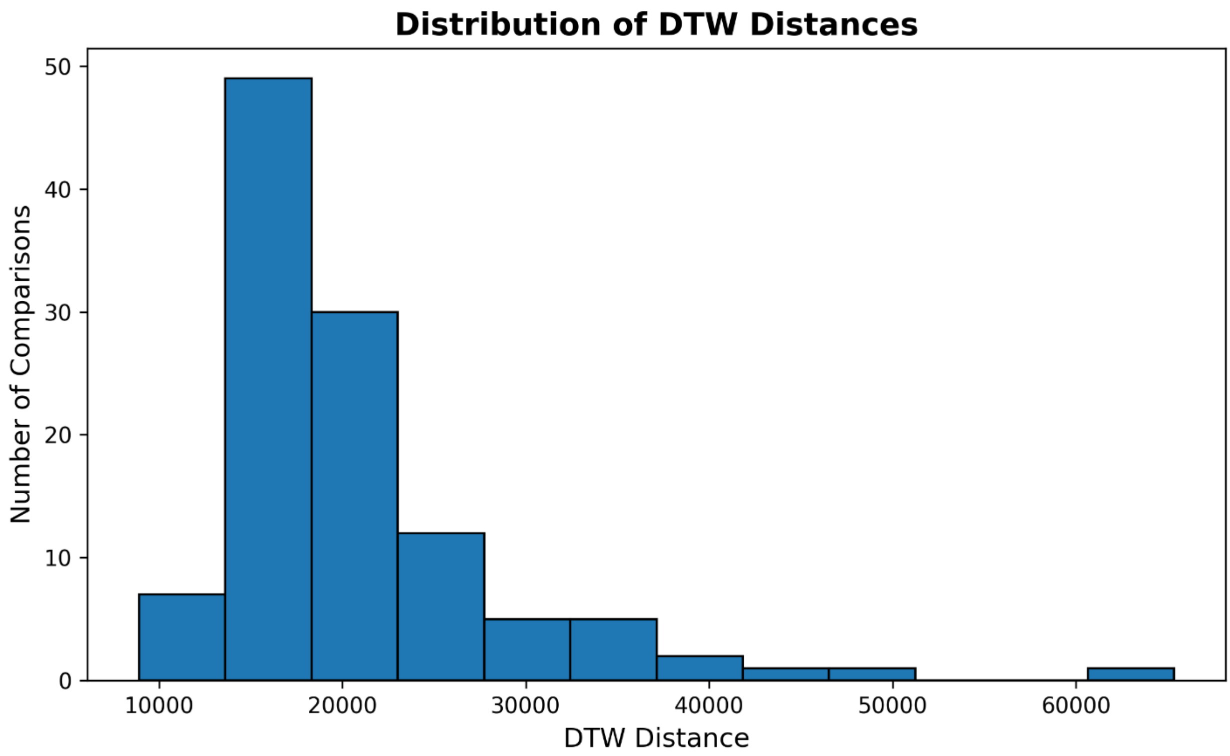

Figure 8.2: Distribution of Dynamic Time Warping (DTW) Distances.

Figure 8.2 presents the distribution of normalised DTW distances across all pronunciation comparisons. Lower DTW values indicate greater acoustic similarity, whereas larger values represent increasing divergence between synthesised and native pronunciations. The observed spread of DTW values demonstrates that pronunciation quality varies substantially across the evaluated street names, confirming that some pronunciations align closely with native speech while others exhibit noticeable acoustic differences.

The DTW distribution complements the PAS results by providing an objective temporal alignment measure instead of just relying solely on feature similarity. DTW has been widely adopted in speech processing because it effectively compensates for differences in speaking rate while preserving meaningful comparisons between speech sequences (Rilliard & De Mareüil, 2011). Consequently, the combined use of PAS and DTW provides a more comprehensive evaluation of pronunciation quality than either metric alone.

The overall results presented in Table 8.1 and Figures 8.1- 8.2 establish that Apple Maps demonstrates moderate pronunciation performance for isiZulu street names. While several pronunciations show relatively close agreement with native speech, substantial variation remains across the dataset, motivating a more detailed street-level analysis presented in the following section.

8.2 Street Level Pronunciation Performance

Following the overall assessment of pronunciation accuracy, a street-level analysis was conducted to identify individual isiZulu street names that exhibited the highest and lowest pronunciation performance. Examining pronunciation accuracy at the lexical level provides greater insight into the specific words that contributed to the overall performance trends observed in the previous section and helps identify recurring pronunciation challenges associated with particular phonetic and linguistic characteristics.

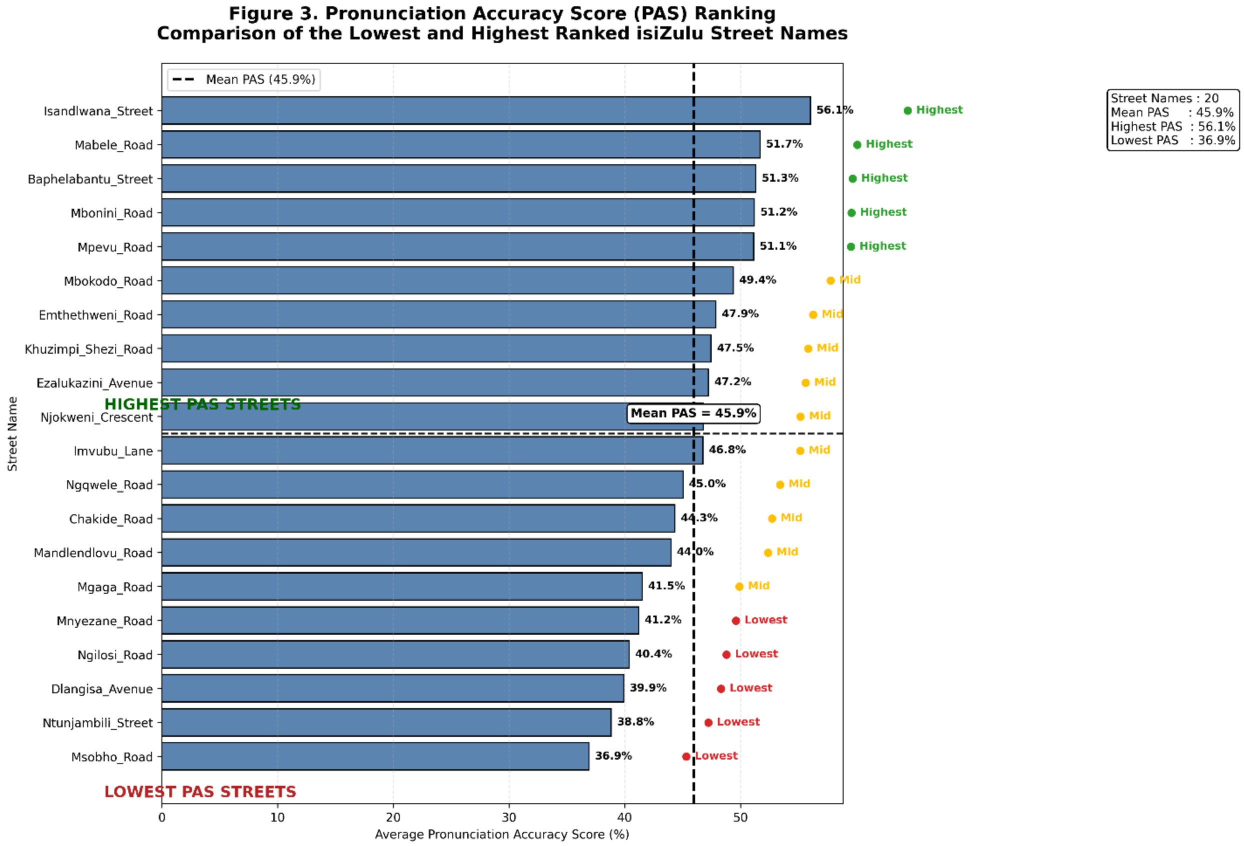

Figure 8.3: Top and Bottom Performing isiZulu Street Names Based on the Pronunciation Accuracy Score (PAS).

Figure 8.3 presents the ten highest-performing and ten lowest-performing isiZulu street names according to their average Pronunciation Accuracy Score (PAS). The results demonstrates that there is a considerable variation in pronunciation performance across individual street names. While several street names achieved relatively higher PAS values, a number of names consistently produced substantially lower scores, indicating greater deviation from native speaker pronunciations.

The poorest-performing street names included streets such as: Ntunjambili Street, Ngilosi Road, Mabele Road, Mqansi Road, Msobho Road, Ngqwele Road, Nkathazo Road, Phezukomkhono Road, Phila Ndwandwe Road, Qhude Road, Sinqandu Road, uMsimbithi Road, uMzimvubu Road, and Ziggaje Road. Manual inspection of the corresponding recordings indicated that these pronunciations frequently contained consonant substitutions, vowel reductions, omission of syllables, and inaccurate realisation of isiZulu click consonants and consonant clusters. These pronunciation deviations contributed directly to lower PAS values and larger acoustic differences from the native speaker recordings.

Recent studies have similarly reported that text-to-speech systems trained primarily on high-resource languages often struggle to accurately synthesise phonologically rich African languages due to limited language-specific resources and insufficient representation of complex phonetic structures during model training (Adebara & Abdul-Mageed, 2022; Bhushan *et al.*, 2025). Consequently, pronunciation errors tend to become more pronounced for words containing uncommon phoneme combinations and language-specific articulatory features.

To further investigate the characteristics of these pronunciation errors, representative observations obtained during manual analysis are summarised in Table 8.2

Table 8.2: Representative Pronunciation Errors Observed in Selected isiZulu Street Names.

Street Name	Manual Listening	PAS (%)	PAS Rating	Comparison	Error Classification/ category	Linguistic Observation
Msobho Road	Poor	36,9	Poor	Consistent	Phonetic substitution	The voiceless alveolar fricative **/s/** was replaced by the voiced alveolar fricative **/z/**, altering the pronunciation of the street name.
Ntunjambili Street	Poor	38,8	Poor	Consistent	Phoneme substitution	The vowel **/u/** was replaced by **/a/** in the first syllable, altering the pronunciation from **Ntunjambili** to **Ntanjambili**.
Dlangisa Avenue	Acceptable	39,9	Poor	Mismatch		Pronunciation considered acceptable
Ngilosi Road	Moderate	40,4	Poor	Mismatch	epenthesis	Apple Maps inserted the vowel /i/ ('iNgilosi')
Mnyezane Road	Poor	41,2	Poor	Consistent	Initial consonant cluster deletion	The initial consonant cluster **/mn/** was omitted, and the final syllable was altered, resulting in a substantially simplified pronunciation (**Mnyezane → Yezain**)
Mgaga Road	Acceptable	41,5	Poor	Mismatch		Pronunciation considered acceptable
Mandlendlovu Road	Acceptable	44	Poor	Mismatch		Pronunciation considered acceptable
Chakide Road	Poor	44,3	Poor	Consistent	Phoneme substitution (vowel and consonant realisation)	The initial **"ch"** was realised as **"tsh"**, while the vowels were altered, producing **"Tshakayd"** instead of **"Chakide**

Street Name	Manual Listening	PAS (%)	PAS Rating	Comparison	Error Classification/ category	Linguistic Observation
Ngqwele Road	Poor	45	Moderate	Mismatch	Consonant cluster simplification	Correctly identified as poor by PAS
Imvubu Lane	Moderate	46,8	Moderate	Consistent		Pronunciation considered acceptable
Njokweni Crescent	Moderate	46,8	Moderate	Consistent		Pronunciation considered acceptable
Ezalukazini Avenue	Moderate	47,2	Moderate	Consistent		Pronunciation considered acceptable
Khuzimpi Shezi Road	Acceptable	47,5	Moderate	Mismatch		Pronunciation considered acceptable
Emthethweni Road	Acceptable	47,9	Moderate	Mismatch		Pronunciation considered acceptable
Mbokodo Road	Acceptable	49,4	Moderate	Mismatch		Pronunciation considered acceptable
Mpevu Road	Moderate	51,1	Moderate	Consistent	Vowel substitution	The vowel **/e/** was replaced by **/i/**, changing the pronunciation from **Mpevu** to **Mpivu** while preserving the surrounding consonant structure.
Mbonini Road	Acceptable	51,2	Moderate	Mismatch		Pronunciation considered acceptable
Baphelabantu Street	Poor	51,3	Moderate	Mismatch	Click consonant substitution	Incorrect syllable production /fe/ ('Bafelabantu')
Mabele Road	Poor	51,7	Moderate	Mismatch	Syllable deletion	Apple Maps omitted the vowel /e/ ('Mable')
Isandlwana Street	Acceptable	56,1	Moderate	Mismatch		Pronunciation considered acceptable

Table 8.2 summarises representative pronunciation patterns identified during the manual comparison of Apple Maps pronunciations against the corresponding native speaker recordings. The observations indicate that pronunciation inaccuracies were not uniformly distributed across the dataset but instead occurred more frequently in street names containing complex phonological structures. Common errors included the omission of syllables, inaccurate production of click consonants, simplification of consonant clusters, vowel reduction, and altered stress placement.

Although the quantitative PAS values provide an objective measure of pronunciation similarity, the qualitative observations presented in Table 8.2 explain the linguistic factors responsible for many of the lower scores. This combination of objective acoustic evaluation and manual phonetic analysis provides a more comprehensive understanding of pronunciation quality than numerical metrics alone. Similar evaluation strategies have been recommended for speech technologies developed for under-resourced languages, where objective acoustic measures are often complemented by linguistic error analysis to better understand model behaviour and identify opportunities for improvement (Adebara *et al.*, 2023; Al-Rami & Zrekat, 2023).

The overall street-level analysis confirms that pronunciation performance varies significantly across isiZulu street names and that many of the lowest-performing examples share common linguistic characteristics that remain challenging for current text-to-speech systems. These findings motivate a further investigation into the relationship between pronunciation accuracy and acoustic similarity, presented in the following section.

Within this category, vowel and consonant substitutions occurred more frequently than click consonant substitutions, suggesting that Apple Maps experienced greater difficulty preserving the phonetic identity of individual speech sounds than maintaining overall word structure. Deletion errors occurred less frequently but were primarily associated with syllable omission and the simplification of initial consonant clusters. Only a small number of insertion errors were identified, indicating that the pronunciation system more commonly simplified or altered existing sounds than introduced additional phonetic segments. These findings suggest that the pronunciation errors were predominantly influenced by the complex phonological characteristics of isiZulu rather than random synthesis variability.

8.3 Relationship Between Pronunciation Accuracy and Dynamic Time Warping

To further investigate the reliability of the Pronunciation Accuracy Score (PAS), the relationship between PAS and Dynamic Time Warping (DTW) was examined. While PAS provides an overall pronunciation accuracy measure derived from multiple acoustic features, DTW directly quantifies the temporal alignment between synthesised and native speech signals. Analysing the relationship between these two measures provides additional evidence regarding the consistency and validity of the proposed evaluation approach.

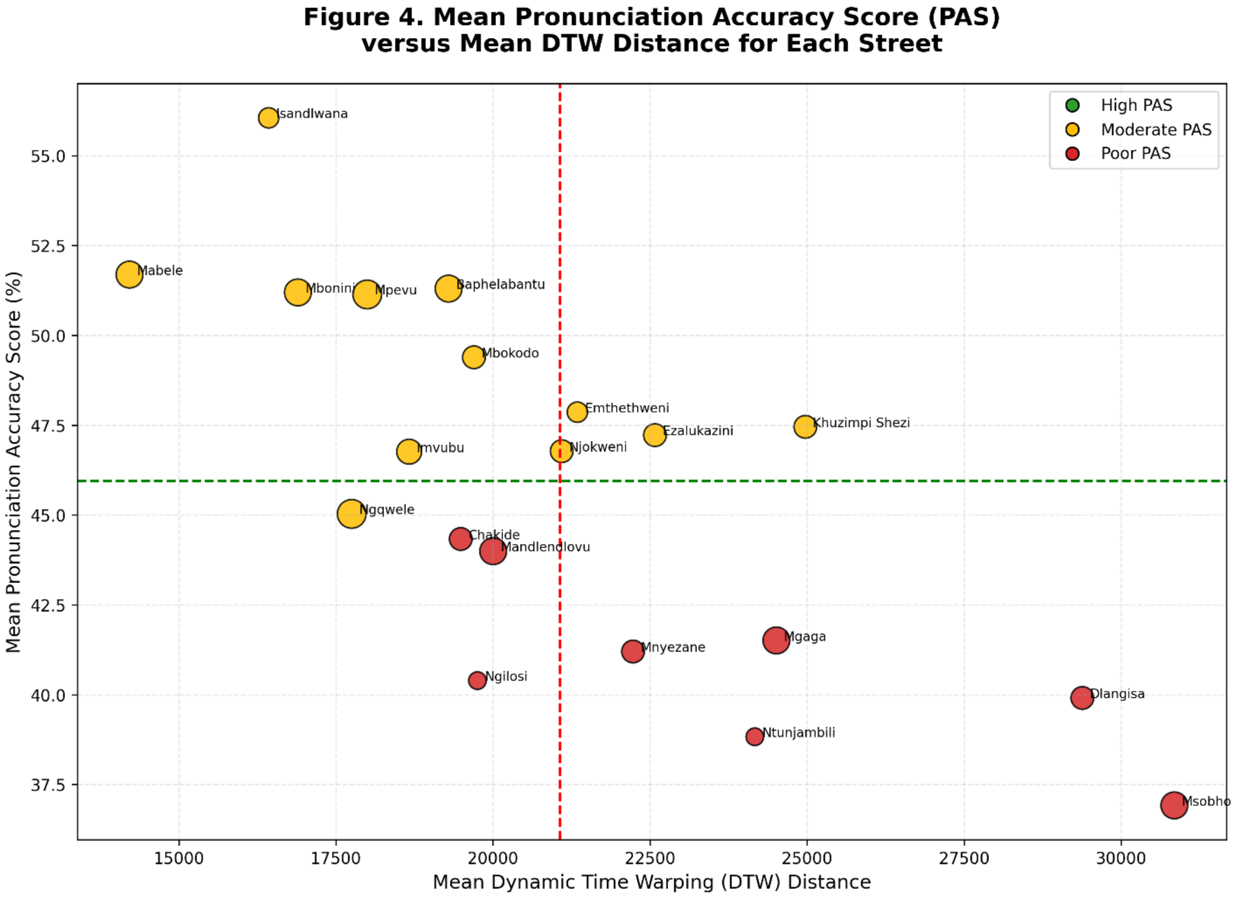

Figure 8.4:. Relationship Between Pronunciation Accuracy Score (PAS) and Dynamic Time Warping (DTW).

Figure 8.4 illustrates the relationship between PAS and normalised DTW values obtained from all pronunciation comparisons. A clear inverse relationship can be observed, whereby higher Pronunciation Accuracy Scores generally correspond to lower DTW values, while lower PAS values are associated with larger DTW distances. This behaviour is expected because a smaller DTW distance indicates closer temporal alignment between Apple Maps pronunciations and native speaker recordings, reflecting greater pronunciation similarity.

The observed relationship demonstrates that both evaluation measures consistently capture pronunciation quality from complementary perspectives. Whereas PAS provides a normalised measure of overall pronunciation similarity, DTW evaluates the temporal correspondence between acoustic feature sequences, making it particularly suitable for speech comparison tasks involving variations in speaking rate and pronunciation duration (Rilliard et at., 2011; Singh, Sharma & Sur, 2025). The agreement between these two measures therefore increases confidence in the reliability of the proposed pronunciation evaluation framework.

Although a general negative relationship is evident, several observations deviate from the overall trend. These cases indicate that temporal similarity alone does not fully explain pronunciation quality. For example, a certain pronunciation may exhibit a relatively small DTW distance while still containing phonetic substitutions, incorrect articulation of click consonants, or altered vowel production that may reduce the overall PAS. Conversely, other pronunciations with moderate DTW values achieved comparatively higher PAS values because the dominant phonetic characteristics were preserved despite minor temporal differences. Similar findings have been reported in recent speech evaluation studies, where multiple complementary acoustic measures were recommended to obtain a more comprehensive assessment of pronunciation quality than any individual metric alone (Kheir, Ali & Chowdhury, 2023; Do, Lee & Lee, 2024).

The results therefore suggest that PAS and DTW should not be interpreted as competing evaluation measures but rather as complementary indicators of pronunciation performance. While DTW effectively measures temporal acoustic similarity, PAS incorporates additional acoustic characteristics that better reflect the overall resemblance between synthesised pronunciations and native isiZulu speech.

Collectively, these findings support the use of a multi-metric evaluation framework for analysing pronunciation quality in low-resource languages. The complementary behaviour of PAS and DTW provides a more robust assessment of synthesised speech than relying solely on a single similarity measure. The following section extends this analysis by examining individual acoustic characteristics, including pitch, pronunciation duration, and speech energy, to identify specific factors contributing to the observed pronunciation differences.

8.4 Acoustic Feature Analysis

Following the evaluation of overall pronunciation accuracy and acoustic similarity, additional analyses were performed to examine individual acoustic characteristics contributing to pronunciation differences between Apple Maps and native speaker recordings. Specifically, average pitch, pronunciation duration, and speech energy were compared to determine whether the synthesised pronunciations preserved the fundamental acoustic properties of native isiZulu speech.

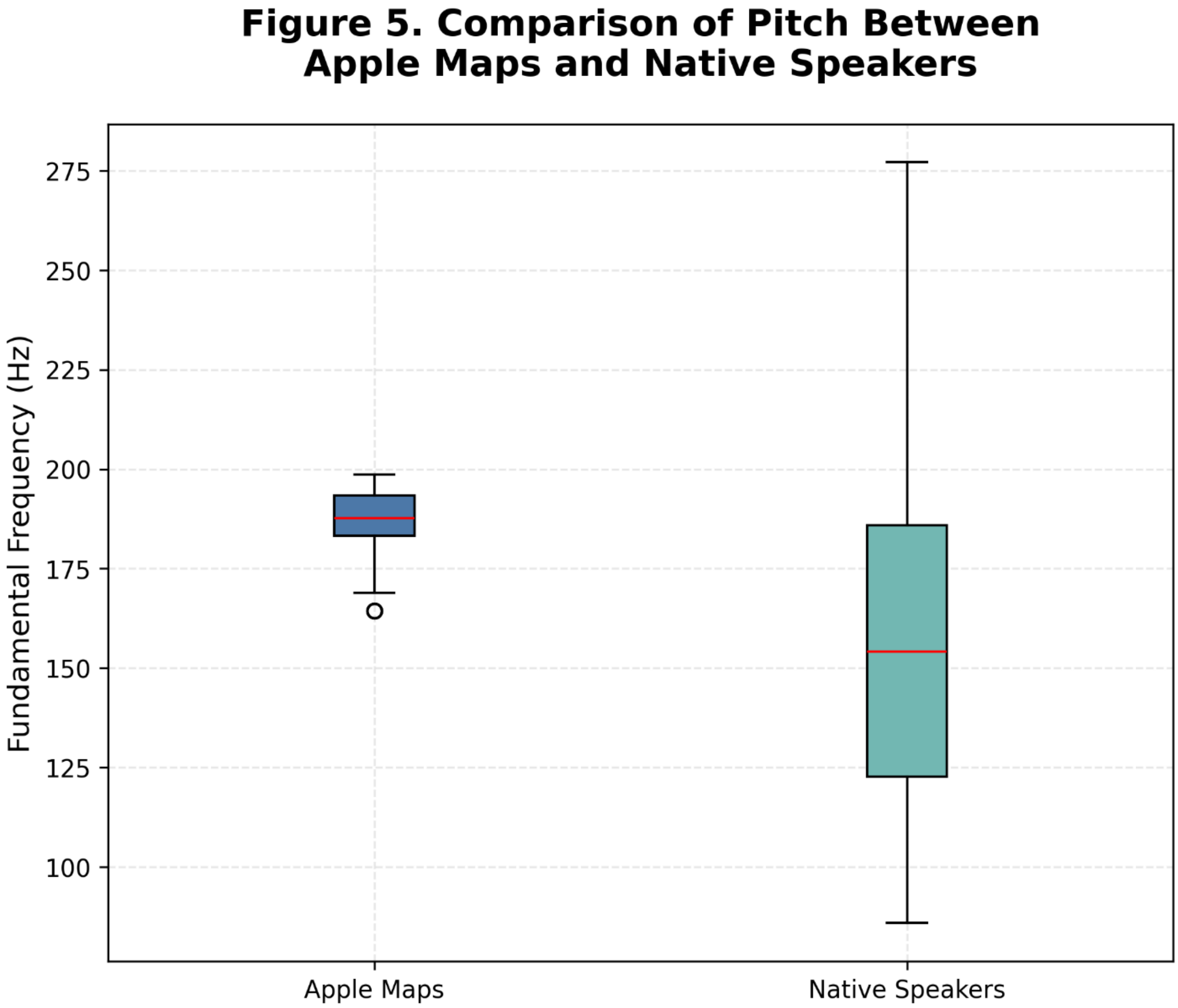

Figure 8.5: Comparison of Average Pitch Between Apple Maps and Native Speaker Recordings.

Figure 8.5 compares the average pitch values extracted from Apple Maps pronunciations and the corresponding native speaker recordings. The results depict clear variations between the synthesised and native pronunciations, with Apple Maps frequently producing pitch values that differed from those that were observed in the native speech. Since pitch contributes to speech naturalness, emphasis, and prosodic variation, these differences suggest that the synthesised pronunciations do not consistently reproduce the prosodic characteristics of native isiZulu speakers.

Recent studies have highlighted that preserving prosodic information remains one of the principal challenges in text-to-speech systems developed for under-resourced languages, particularly where limited speech corpora constrain the learning of language-specific intonation patterns (Adebara et al., 2023; Geng et al., 2025). The pitch differences observed in this study are therefore consistent with challenges previously reported for African language speech synthesis.

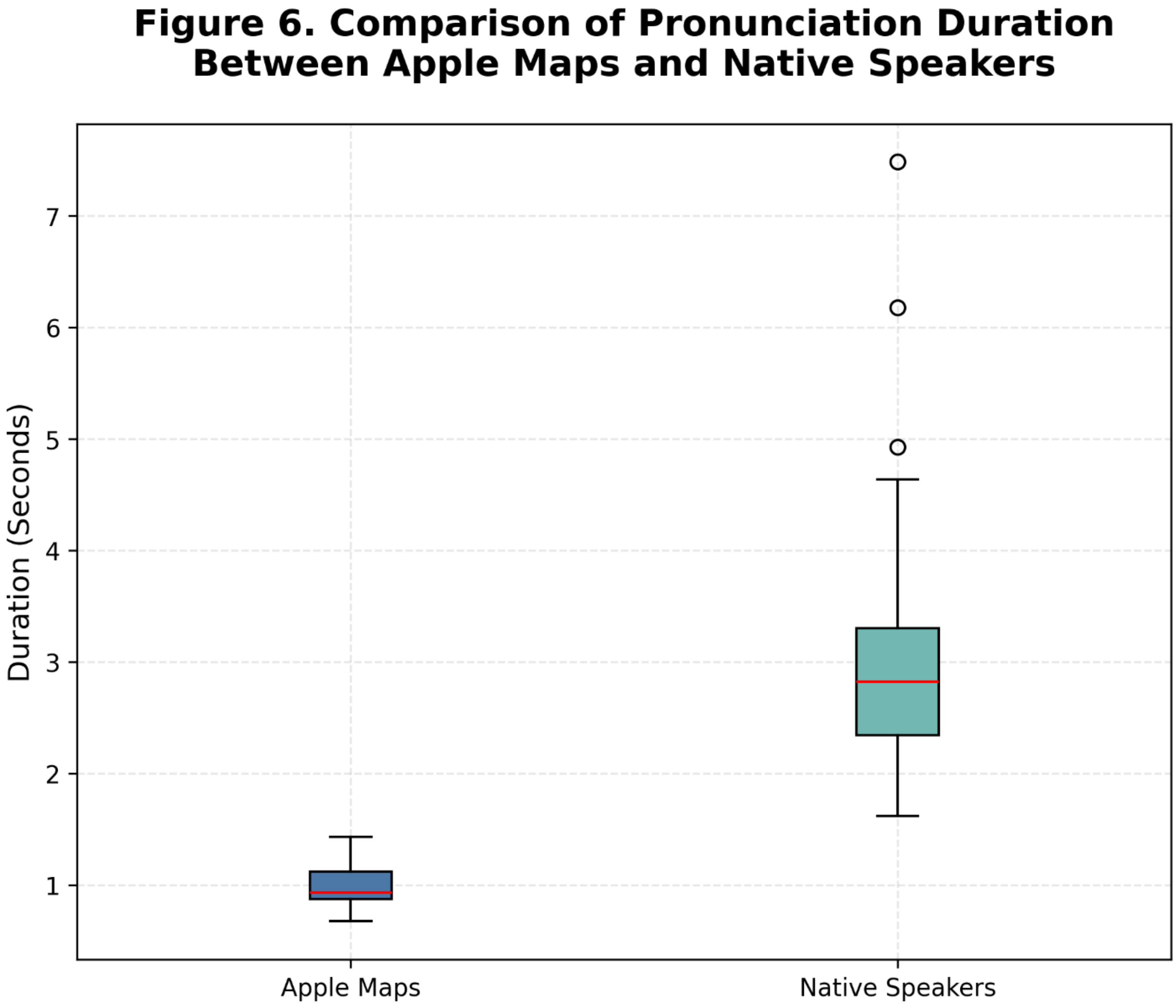

Figure 8.6: Comparison of Pronunciation Duration Between Apple Maps and Native Speaker Recordings.

Pronunciation duration provides an indication of speaking rate and temporal characteristics during speech production. Figure 8.6 demonstrates that Apple Maps generally produced shorter pronunciation durations than the corresponding native speaker recordings. While shorter durations do not necessarily indicate incorrect pronunciation, substantial differences may influence articulation quality and reduce speech naturalness, particularly for agglutinative languages such as isiZulu where complete pronunciation of syllables contributes to intelligibility.

The consistently shorter durations observed across many street names suggest that Apple Maps may simplify or accelerate portions of the synthesised pronunciations. Similar temporal compression has previously been reported in speech synthesis systems trained using limited language-specific datasets, where insufficient modelling of pronunciation timing results in shortened or incomplete articulation (Kheir *et al.*, 2023; Do *et al.*, 2024).

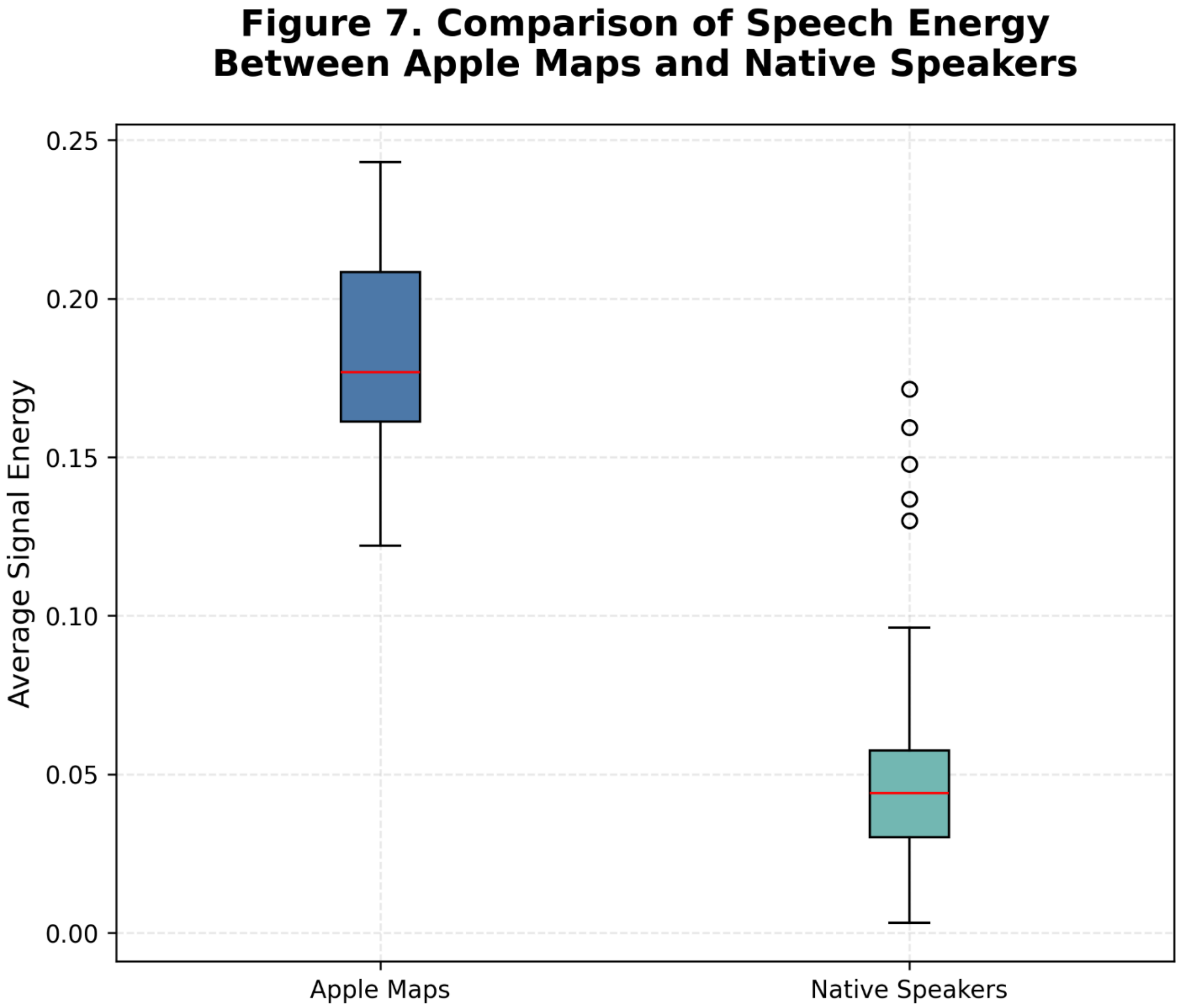

Figure 8.7: Comparison of Speech Energy Between Apple Maps and Native Speaker Recordings.

Speech energy represents the relative intensity of the speech signal and provides additional information regarding pronunciation dynamics. Figure 8.7 indicates measurable differences between the synthesised and native recordings, although these variations appear less pronounced than those observed for pitch and duration. Overall, the synthesised pronunciations generally maintained comparable energy levels, suggesting that speech intensity contributes less to the observed pronunciation inaccuracies than temporal and prosodic characteristics.

This observation implies that pronunciation differences identified in the earlier analyses are more likely associated with phonetic realisation, articulation timing, and prosodic variation rather than differences in speech loudness alone. Previous speech-processing studies similarly report that acoustic similarity measures are more strongly influenced by spectral and temporal characteristics than by overall signal energy, particularly when evaluating synthesised speech against natural recordings (Maryn et al., 2009; Besacier et al., 2014).

The collective analyses presented in Figures 8.5-8.7 demonstrate that Apple Maps does not consistently preserve several important acoustic characteristics of native isiZulu pronunciation. Among the three features examined, the greatest differences were observed in pitch and pronunciation duration, whereas speech energy remained comparatively similar between the synthesised and native recordings. These findings complement the PAS and DTW analyses presented earlier and provide additional evidence that pronunciation inaccuracies arise primarily from differences in prosodic and temporal speech characteristics rather than signal intensity alone. To further illustrate these acoustic differences, the following section presents a qualitative case study using detailed spectrographic and feature-based visualisations of selected isiZulu street names.

8.5 Qualitative Case Study Analysis

While the preceding sections presented quantitative evidence of pronunciation performance across the complete dataset, a qualitative analysis was also conducted to examine how pronunciation differences manifest within individual speech signals. Two representative isiZulu street names, Mabele Road and Mgaga Road, were selected as case studies because they demonstrated pronunciation characteristics representative of the broader patterns observed throughout the dataset. The objective of this analysis was not to evaluate every street individually, but rather to illustrate how the previously identified differences in pronunciation accuracy, temporal alignment, and acoustic features appear within the speech signal itself.

Figure 8.8 presents spectrograms of the selected Apple Maps pronunciations alongside their corresponding native speaker recordings. Spectrograms provide a visual representation of speech by illustrating the distribution of acoustic energy across time and frequency. Visual inspection reveals noticeable differences in spectral structure between the synthesised and native pronunciations. Several phonetic regions exhibit reduced spectral detail in the Apple Maps recordings, while portions of the native pronunciations display richer harmonic patterns and more clearly defined formant structures.

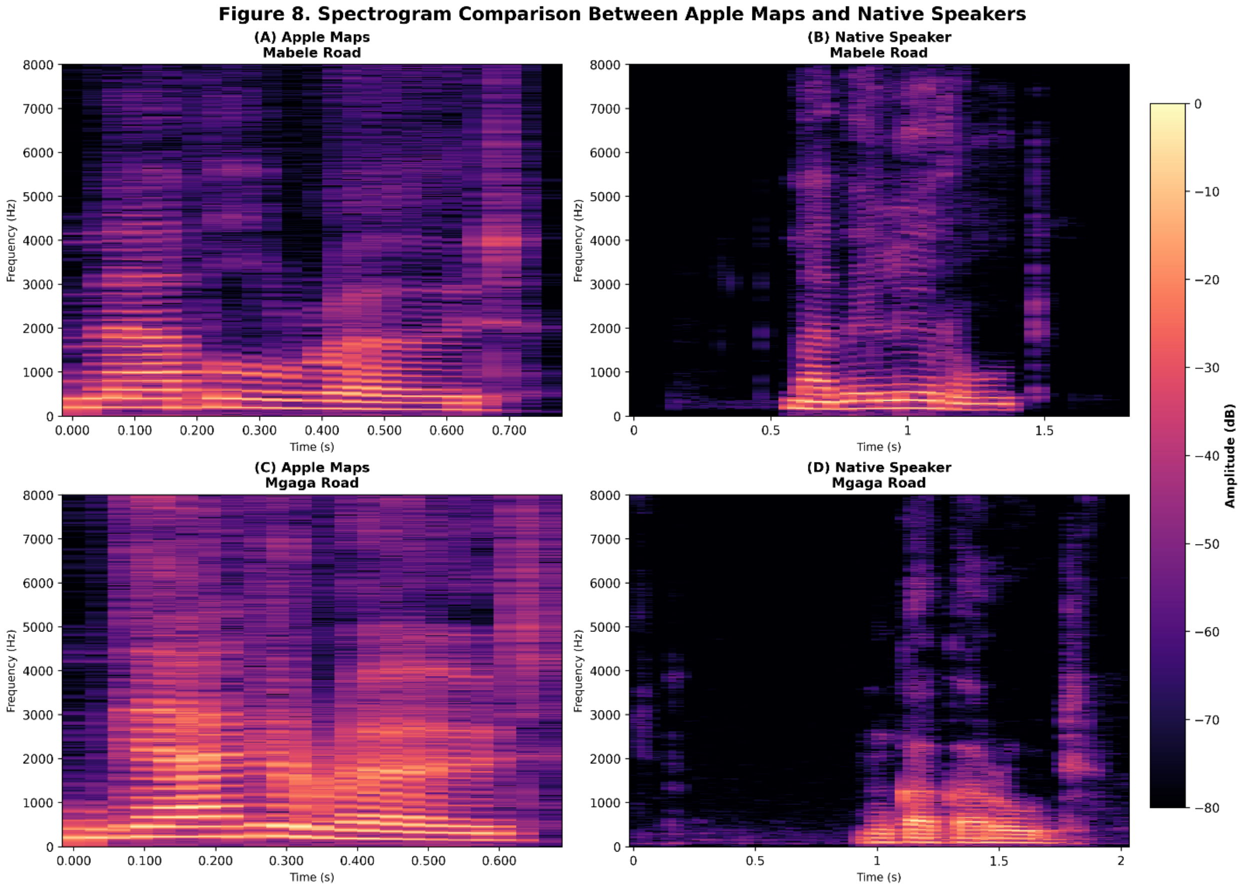

Figure 8.8: Spectrogram Comparison Between Apple Maps and Native Speaker Recordings.

These differences are particularly evident around consonant transitions and vowel regions, where native speakers exhibit smoother spectral continuity than the synthesised speech. Similar observations have been reported in speech synthesis studies for low-resource languages, where insufficient modelling of language-specific phonetic characteristics often results in simplified spectral representations and reduced articulation quality (Adebara et al., 2023; David et al., 2025).

To further examine perceptually relevant acoustic differences, Mel spectrograms were generated for the same pronunciations. Unlike conventional spectrograms, Mel spectrograms represent speech using a frequency scale that more closely reflects human auditory perception. As illustrated in Figure 8.9 below, the native speaker recordings demonstrate smoother and more continuous energy distributions across the Mel frequency bands, whereas Apple Maps pronunciations exhibit comparatively irregular energy patterns within several speech segments.

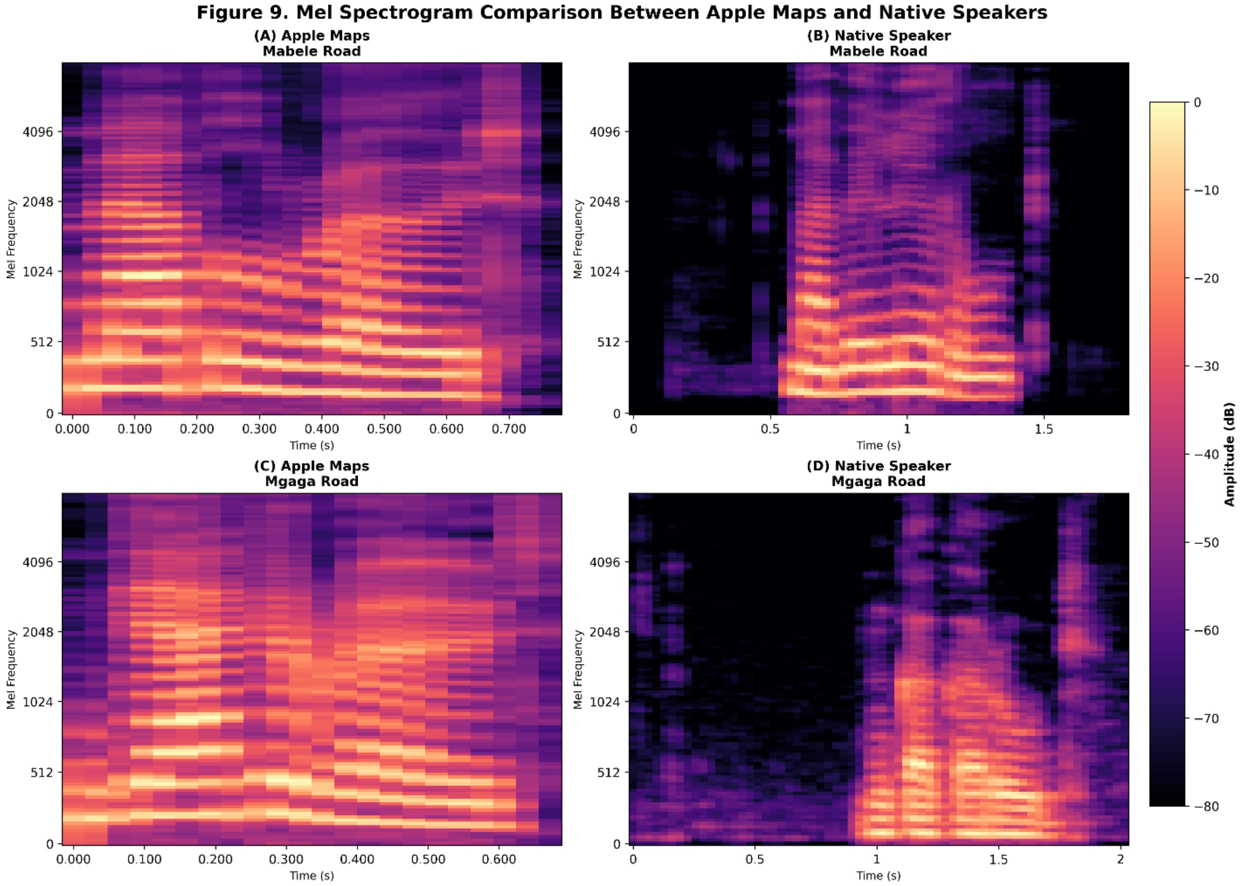

Figure 8.9: Mel Spectrogram Comparison Between Apple Maps and Native Speaker Recordings.

These observations suggest that the synthesised pronunciations do not fully preserve the perceptually important acoustic characteristics associated with natural isiZulu speech. Previous work has shown that Mel-frequency representations are particularly effective for analysing pronunciation quality because they capture spectral properties closely associated with human speech perception (Davis & Mermelstein, 1980; Majeed et al., 2015).

Mel Frequency Cepstral Coefficients (MFCCs) provide a compact representation of the spectral envelope of speech and are widely employed in automatic speech recognition and pronunciation analysis. Figure 8.10 below compares the MFCC heatmaps extracted from the synthesised and native pronunciations. Although several broad spectral characteristics are shared, noticeable differences remain within multiple coefficient regions, indicating variations in the acoustic signatures of the pronunciations.

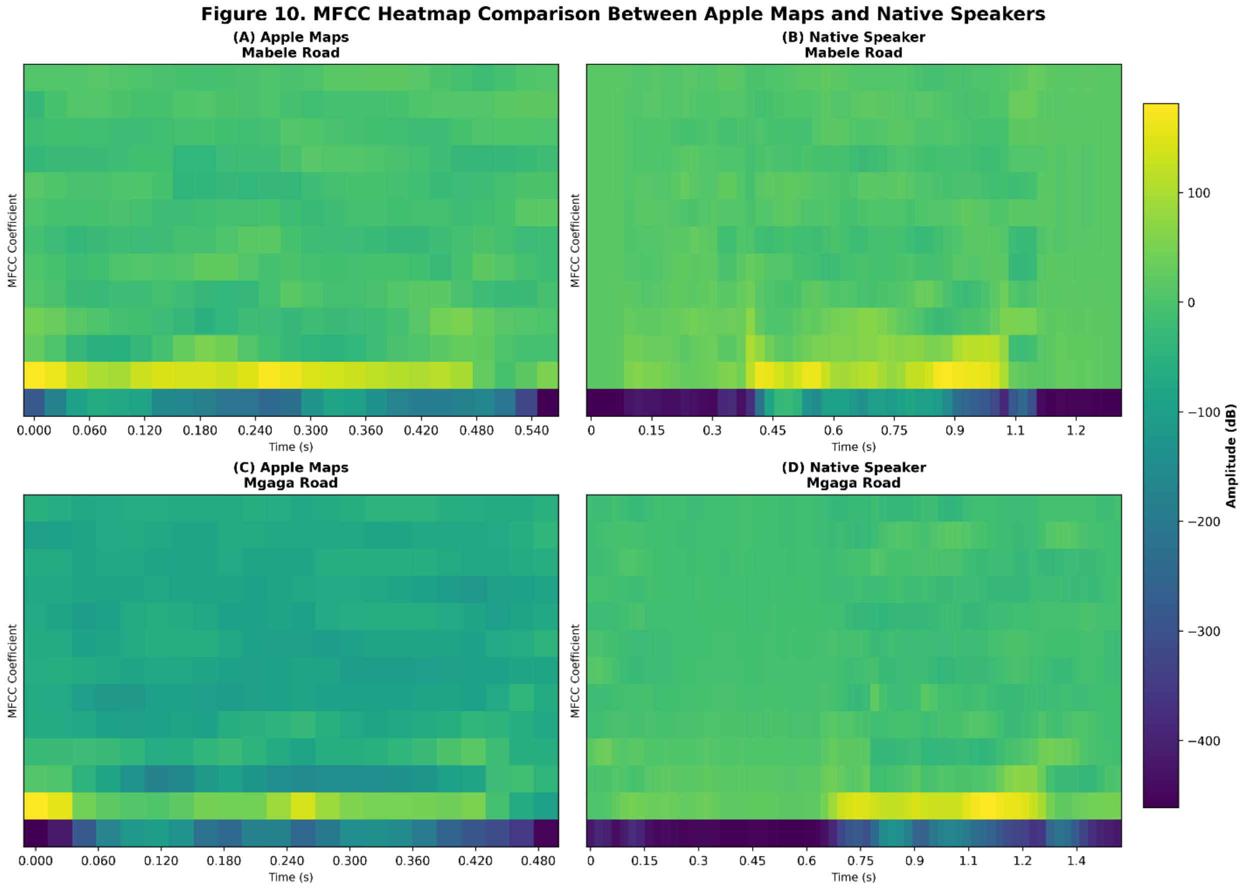

Figure 8.10: MFCC Heatmap Comparison Between Apple Maps and Native Speaker Recordings.

These MFCC differences correspond closely with the PAS and DTW analyses presented earlier, providing additional evidence that Apple Maps does not consistently reproduce the spectral characteristics observed in native isiZulu speech. Because MFCCs represent the features used during the pronunciation comparison process, the observed variations further support the reliability of the proposed evaluation methodology.

Figure 8.11 below illustrates the Dynamic Time Warping alignment generated for the representative case study pronunciations. The alignment path demonstrates how DTW establishes correspondences between the synthesised and native feature sequences despite differences in speaking rate and pronunciation duration. An alignment path closely following the diagonal indicates greater temporal similarity, whereas larger deviations represent increased temporal distortion between the two pronunciations.

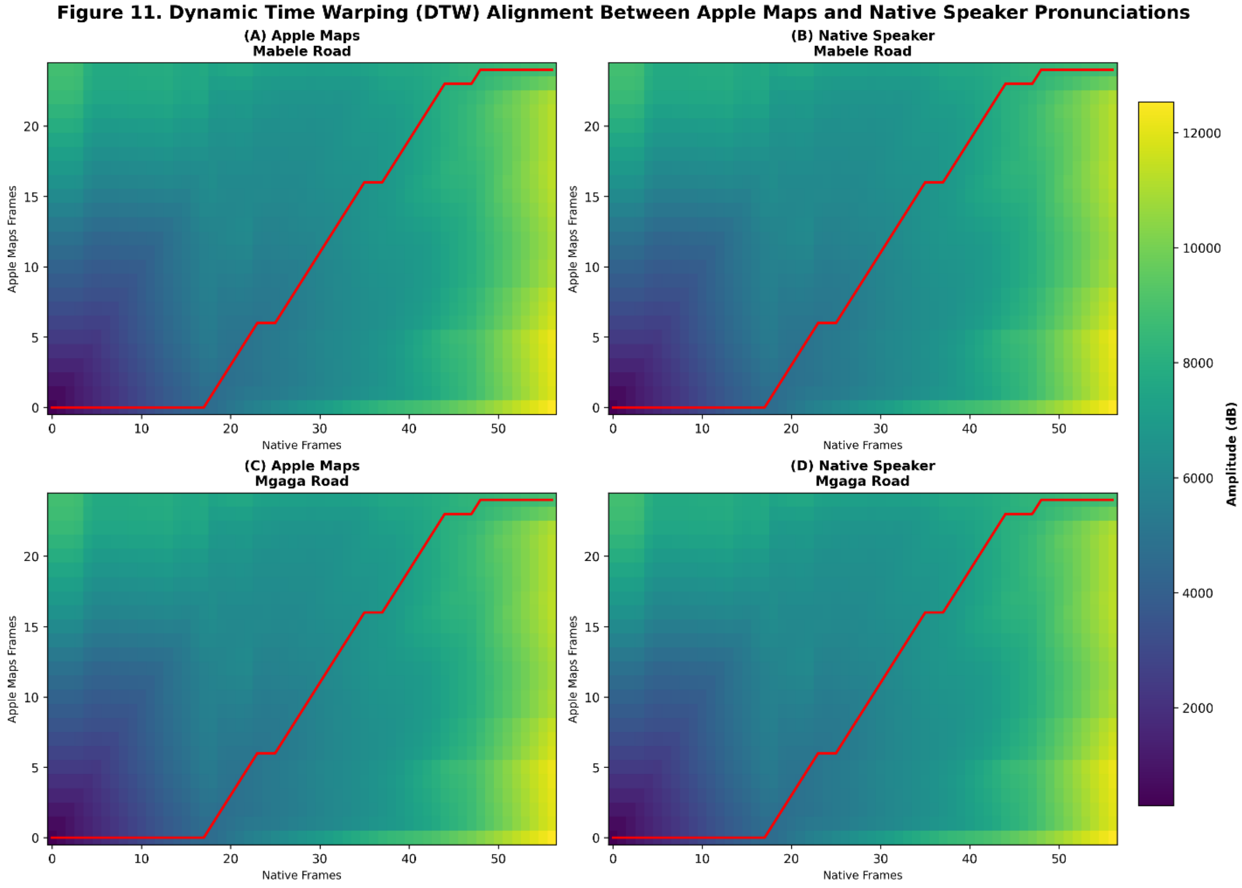

Figure 8.11: Dynamic Time Warping Alignment Between Apple Maps and Native Speaker Pronunciations.

The alignments observed in Figure 8.11 correspond well with the quantitative DTW values reported earlier, demonstrating that larger DTW distances are associated with increasingly complex alignment paths. This visual evidence supports the conclusion that pronunciation differences arise not only from spectral variation but also from differences in temporal articulation. As a result, the qualitative case study reinforces the findings obtained from the objective evaluation metrics and provides an intuitive visual explanation of the pronunciation differences observed throughout the dataset.

In conclusion, the qualitative analyses presented in Figures 8.8-8.11 provide visual confirmation of the quantitative findings discussed throughout this section. The spectrograms, Mel spectrograms, MFCC heatmaps, and DTW alignments consistently demonstrate that Apple Maps pronunciations differ from native isiZulu speech in both spectral and temporal characteristics. Although only two representative street names were examined in detail, the observed acoustic patterns are consistent with the broader pronunciation trends identified across the complete dataset. These findings strengthen the overall validity of the proposed evaluation framework by demonstrating agreement between objective performance metrics and direct acoustic inspection of the speech signals.

8.6 Summary

This section presented the preliminary findings obtained from evaluating Apple Maps pronunciations of isiZulu street names against native speaker recordings using both quantitative and qualitative speech analysis techniques. The results demonstrated that pronunciation performance varied considerably across the analysed dataset, with the overall Pronunciation Accuracy Score (PAS) indicating moderate agreement between synthesised and native pronunciations. Dynamic Time Warping (DTW) analysis further confirmed that temporal acoustic differences remained present across many pronunciation comparisons.

Street-level analysis revealed that pronunciation inaccuracies were concentrated in specific isiZulu street names containing complex phonological structures, including consonant clusters, click consonants, and agglutinative word formations. Manual error analysis showed that the most frequently observed pronunciation errors included phoneme substitutions, consonant cluster simplifications, syllable omissions, and inaccurate articulation of language-specific phonetic features. These findings suggest that current speech synthesis systems continue to experience difficulties modelling the linguistic characteristics of under-resourced African languages.

Additional acoustic analyses demonstrated measurable differences between Apple Maps and native speaker recordings in terms of pitch, pronunciation duration, and spectral characteristics. The qualitative case studies further illustrated these differences through spectrograms, Mel spectrograms, MFCC heatmaps, and Dynamic Time Warping alignments, providing visual confirmation of the quantitative evaluation results. Collectively, the findings presented in this chapter establish a comprehensive baseline assessment of pronunciation performance and provide the foundation for the discussion and interpretation of the results presented in the following chapter.

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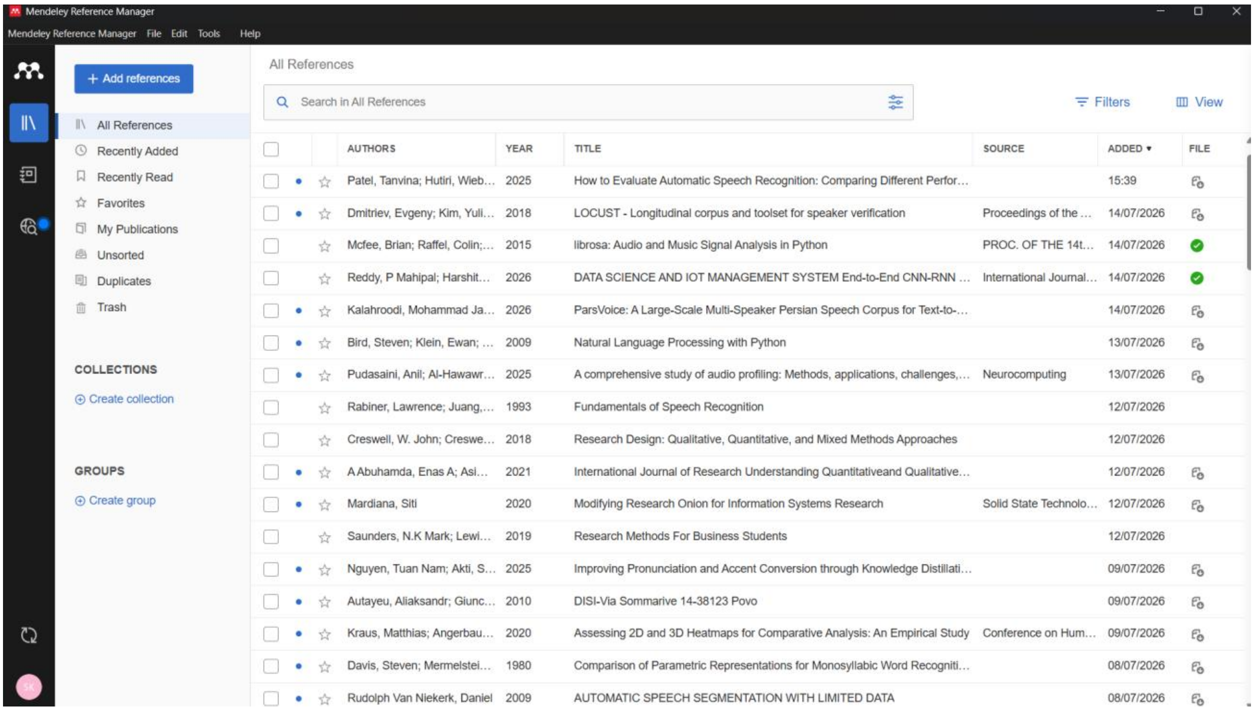
10 Appendix

10.1 A: Screenshot of referencing manager home page



10.2 B: Screenshot of Signed plagiarism

Plagiarism Pledge

1. I have read Unisa's plagiarism policy.
2. I understand Unisa's plagiarism policy.
3. I agree to abide by Unisa's plagiarism policy.
4. I have read the direct copying, plagiarism, and "patch-writing" document.
5. I understand what direct copying, plagiarism, and "patch-writing" is.
6. I undertake to avoid copying directly, plagiarism and patch writing.
7. All academic work, written or otherwise, that I submit is expected to be the result of my own skill and labour.
8. I understand that, if I am guilty of the infringement of breach of copyright/plagiarism or unethical practice, I will be subject to the applicable disciplinary code as determined by Unisa.
9. I understand that it is my responsibility to use Turnitin (or similar plagiarism tool) to check submitted research for direct copying or plagiarism.
10. The supervisor has the right and responsibility to return any work to be revised if plagiarism is detected.

Student Name: Sam Khumalo

Student Number: 1188-099-6

Student Signature:

Date: 20-07-2026

10.3 C: Turnitin Receipt

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10.4 D: Supplementary Material

The Python implementation, experimental scripts, generated figures, and supporting files developed during this research are maintained in a GitHub repository to promote transparency and reproducibility.

Repository:

https://github.com/Dot-Trojan-Horse/NLP-Research-Project